Plain vs advection-corrected — samples 1-4 of 100

plain sum (max 8.6 mm)

advection-corrected (max 3.8 mm)

corrected - plain (mm)

Euler characteristic χ

random
2024-01-22 05:00
60.7N 4.8E

plain
corrected

threshold (mm in 1 h)

plain sum (max 0.9 mm)

advection-corrected (max 0.7 mm)

corrected - plain (mm)

Euler characteristic χ

random
2017-11-02 07:00
47.0N 9.6E

threshold (mm in 1 h)

plain sum (max 0.2 mm)

advection-corrected (max 0.1 mm)

corrected - plain (mm)

Euler characteristic χ

random
2013-11-05 21:00
60.5N 18.8E

mm in 1 h (log)

difference (mm)

threshold (mm in 1 h)

plain sum (max 0.1 mm)

advection-corrected (max 0.1 mm)

corrected - plain (mm)

Euler characteristic χ

random
2013-05-12 20:00
36.5N -4.5E

threshold (mm in 1 h)

Plain vs advection-corrected — samples 5-8 of 100

plain sum (max 7.9 mm)
advection-corrected (max 4.2 mm)
corrected - plain (mm)
Euler characteristic χ
random 2023-01-08 04:00 50.6N -5.0E
plain
corrected
threshold (mm in 1 h)

plain sum (max 70.9 mm)
advection-corrected (max 38.9 mm)
corrected - plain (mm)
Euler characteristic χ
random 2025-07-19 10:00 46.9N 6.3E
mm in 1 h (log)
difference (mm)
threshold (mm in 1 h)

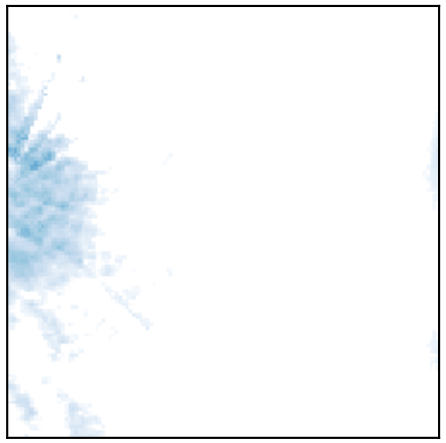
plain sum (max 0.0 mm)
advection-corrected (max 0.0 mm)
corrected - plain (mm)
Euler characteristic χ
random 2023-07-07 13:00 53.9N 9.5E
threshold (mm in 1 h)

plain sum (max 0.7 mm)
advection-corrected (max 0.5 mm)
corrected - plain (mm)
Euler characteristic χ
random 2022-12-21 01:00 58.5N 9.5E
threshold (mm in 1 h)

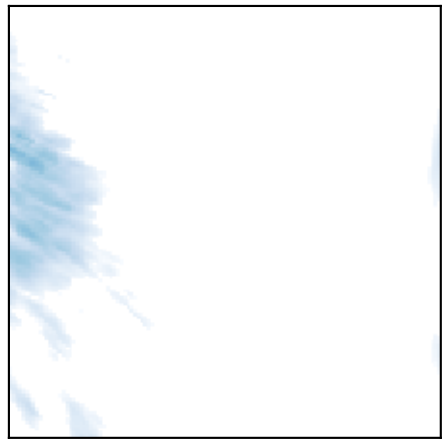
Plain vs advection-corrected — samples 9-12 of 100

random
2022-05-14 08:00
65.3N 14.8E

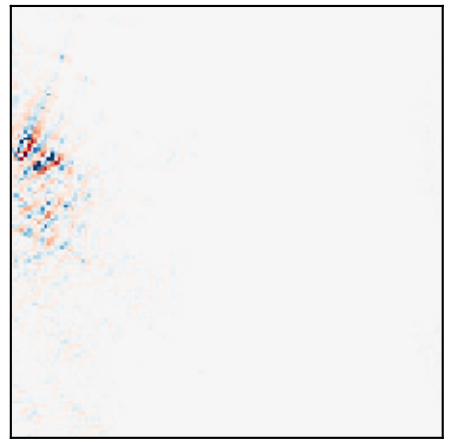
plain sum (max 3.6 mm)



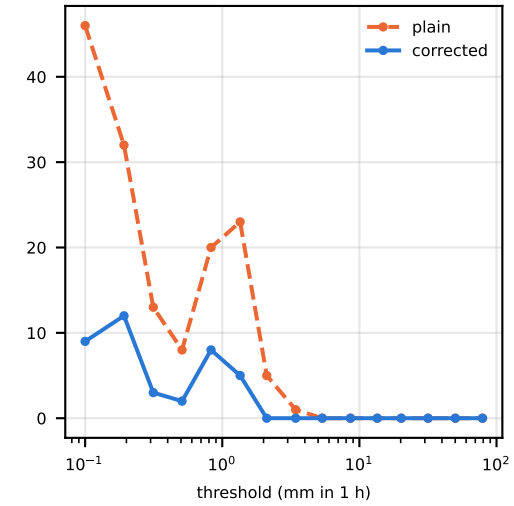
advection-corrected (max 2.0 mm)



corrected - plain (mm)

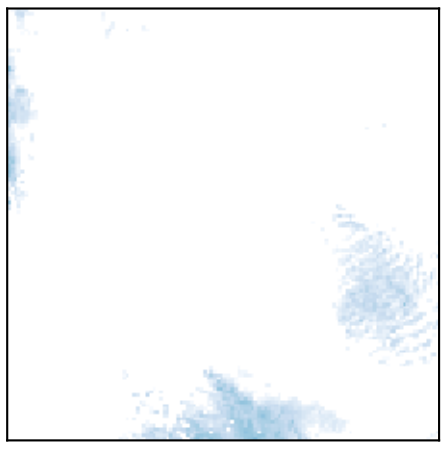


Euler characteristic χ

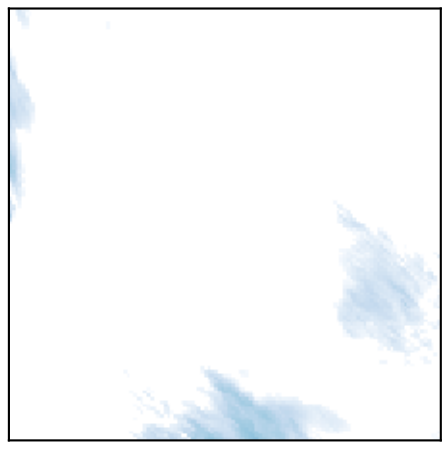


random
2025-11-15 16:00
49.3N 9.6E

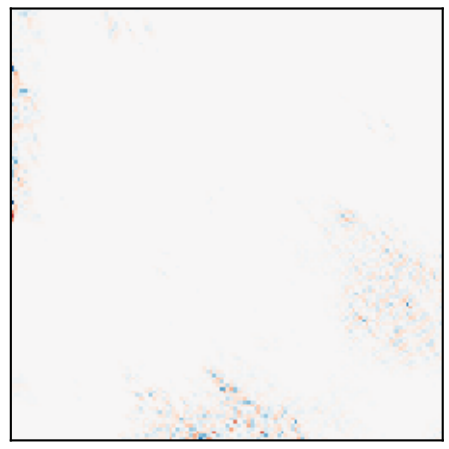
plain sum (max 1.8 mm)



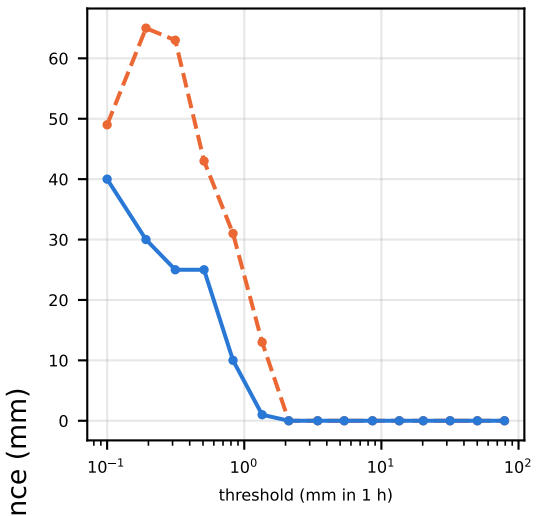
advection-corrected (max 1.6 mm)



corrected - plain (mm)

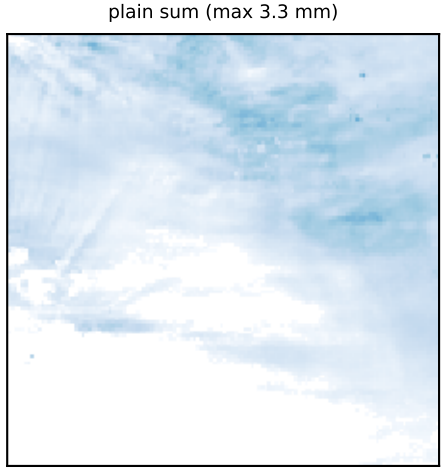


Euler characteristic χ

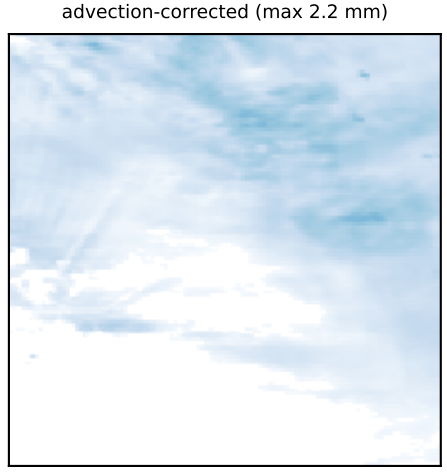


random
2023-04-28 01:00
57.8N -3.5E

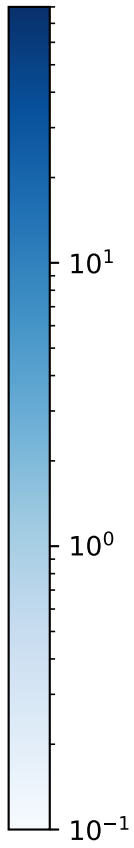
plain sum (max 3.3 mm)



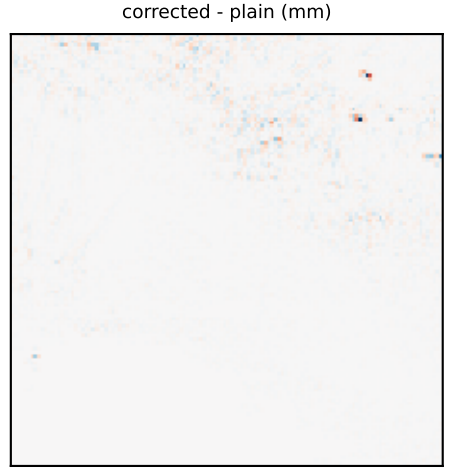
advection-corrected (max 2.2 mm)



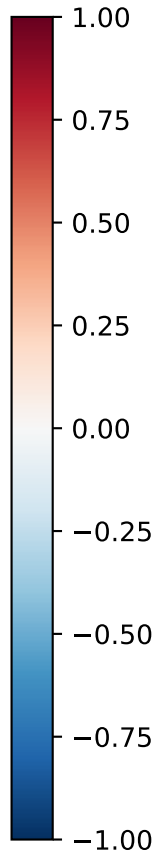
mm in 1 h (log)



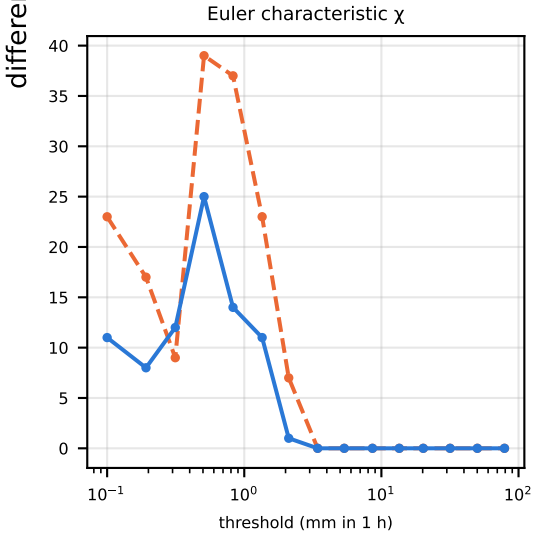
corrected - plain (mm)



difference (mm)

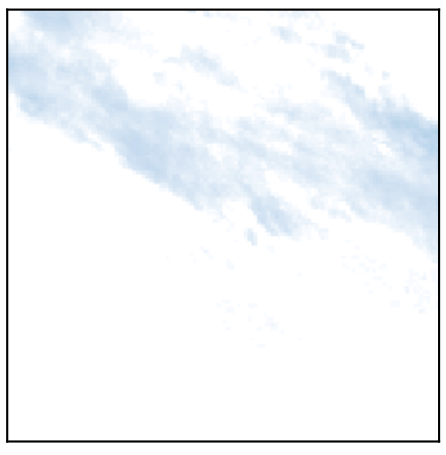


Euler characteristic χ

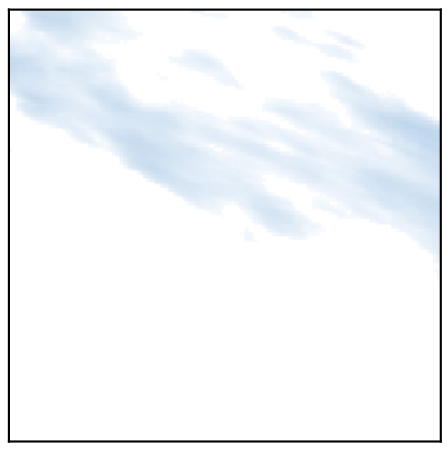


random
2012-10-07 01:00
51.1N 20.6E

plain sum (max 0.8 mm)



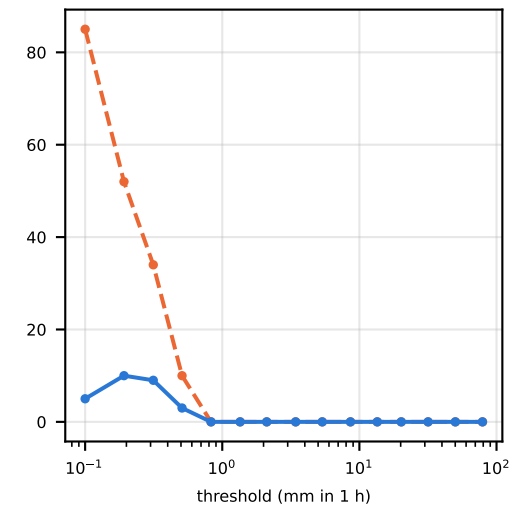
advection-corrected (max 0.6 mm)



corrected - plain (mm)



Euler characteristic χ



Plain vs advection-corrected — samples 13-16 of 100

plain sum (max 27.3 mm)
advection-corrected (max 14.5 mm)
corrected - plain (mm)
Euler characteristic χ
random 2024-10-08 15:00 53.3N -2.0E
plain
corrected
threshold (mm in 1 h)

plain sum (max 1.1 mm)
advection-corrected (max 1.1 mm)
corrected - plain (mm)
Euler characteristic χ
random 2013-09-10 21:00 46.2N 22.9E
mm in 1 h (log)
difference (mm)
threshold (mm in 1 h)

plain sum (max 3.3 mm)
advection-corrected (max 1.3 mm)
corrected - plain (mm)
Euler characteristic χ
random 2022-12-21 18:00 46.8N 16.3E
threshold (mm in 1 h)

plain sum (max 1.1 mm)
advection-corrected (max 1.1 mm)
corrected - plain (mm)
Euler characteristic χ
random 2016-03-17 14:00 46.5N -0.4E
threshold (mm in 1 h)

Plain vs advection-corrected — samples 17-20 of 100

plain sum (max 0.0 mm)

advection-corrected (max 0.0 mm)

corrected - plain (mm)

Euler characteristic χ

random
2024-11-06 05:00
51.6N 9.6E

plain
corrected

threshold (mm in 1 h)

plain sum (max 3.1 mm)

advection-corrected (max 2.2 mm)

corrected - plain (mm)

Euler characteristic χ

random
2022-05-04 14:00
46.5N -0.4E

mm in 1 h (log)

difference (mm)

threshold (mm in 1 h)

plain sum (max 6.5 mm)

advection-corrected (max 2.0 mm)

corrected - plain (mm)

Euler characteristic χ

random
2018-04-03 12:00
55.9N 17.7E

threshold (mm in 1 h)

plain sum (max 0.1 mm)

advection-corrected (max 0.1 mm)

corrected - plain (mm)

Euler characteristic χ

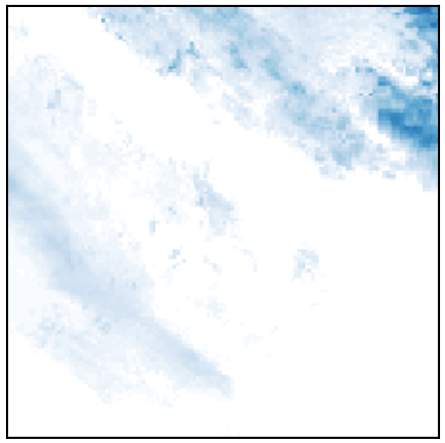
random
2020-04-30 07:00
60.7N 14.1E

threshold (mm in 1 h)

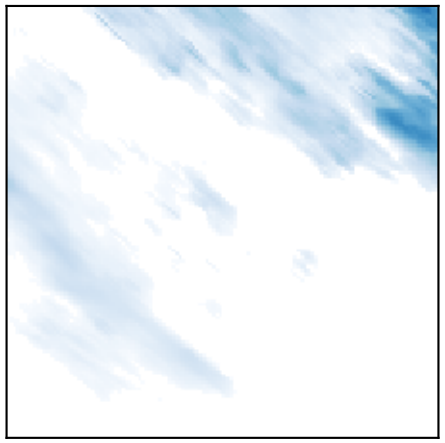
Plain vs advection-corrected — samples 21-24 of 100

random
2013-08-14 21:00
48.4N 23.5E

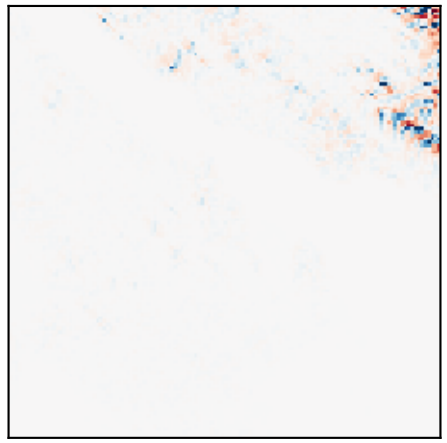
plain sum (max 16.6 mm)



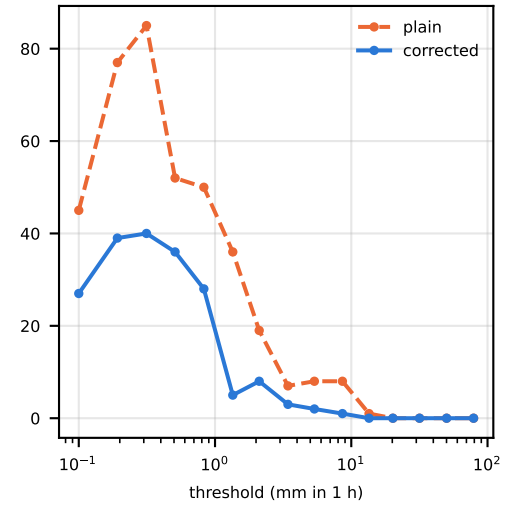
advection-corrected (max 13.3 mm)



corrected - plain (mm)

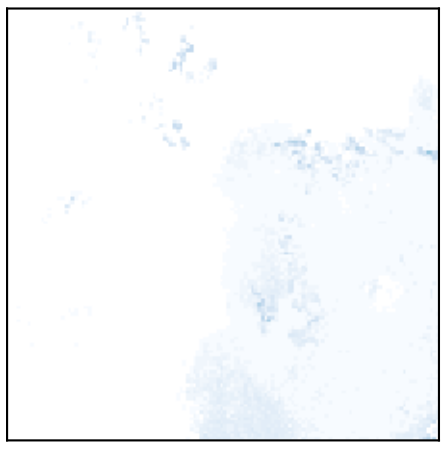


Euler characteristic χ

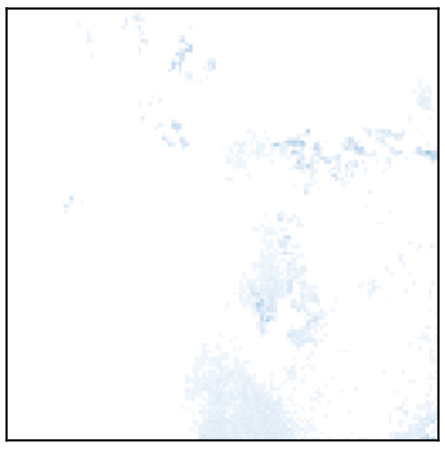


random
2015-05-12 01:00
46.2N 22.9E

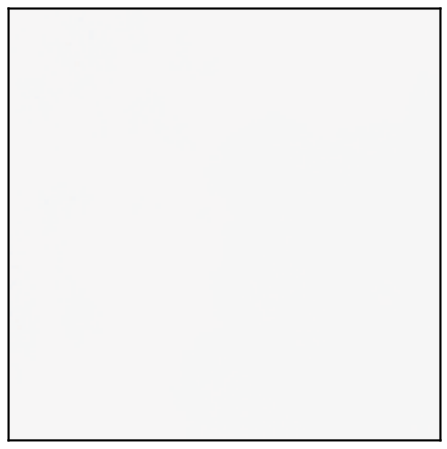
plain sum (max 1.4 mm)



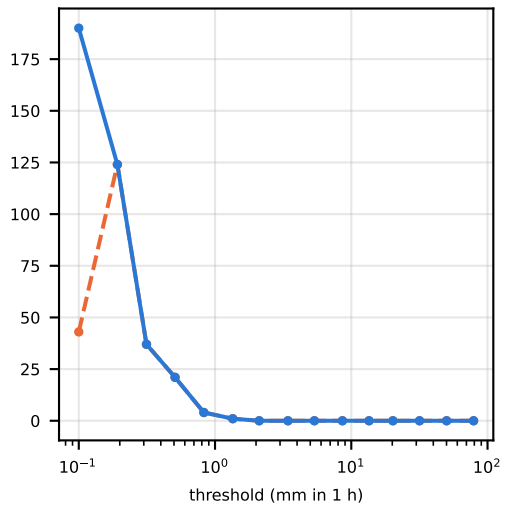
advection-corrected (max 1.4 mm)



corrected - plain (mm)

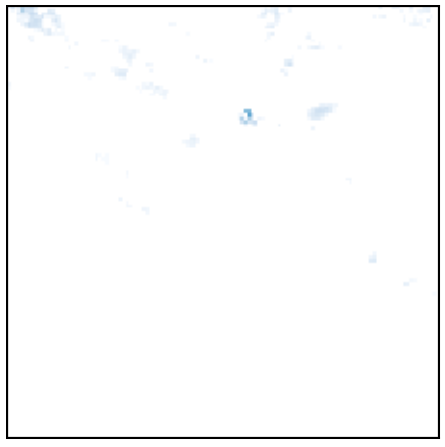


Euler characteristic χ



random
2016-12-16 19:00
40.5N -8.6E

plain sum (max 3.3 mm)



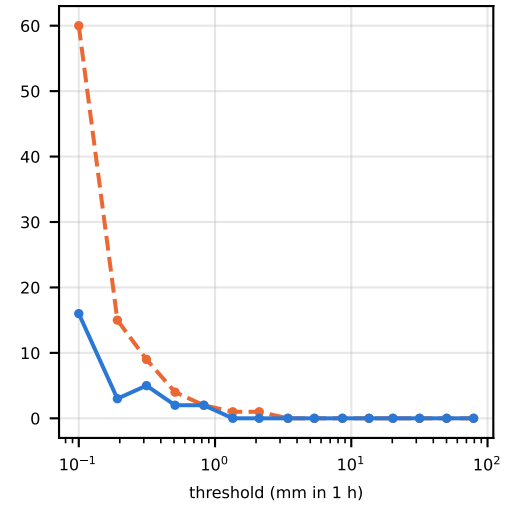
advection-corrected (max 1.2 mm)



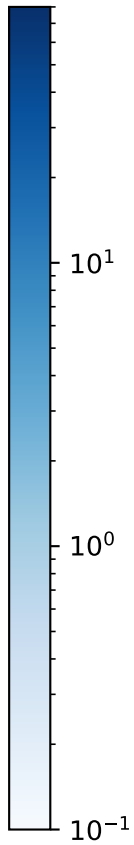
corrected - plain (mm)



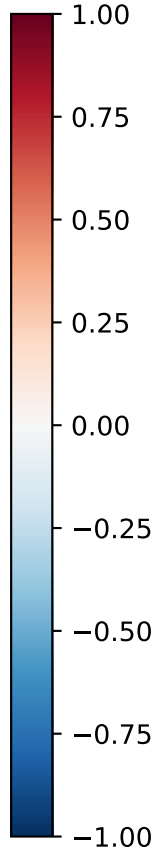
Euler characteristic χ



mm in 1 h (log)

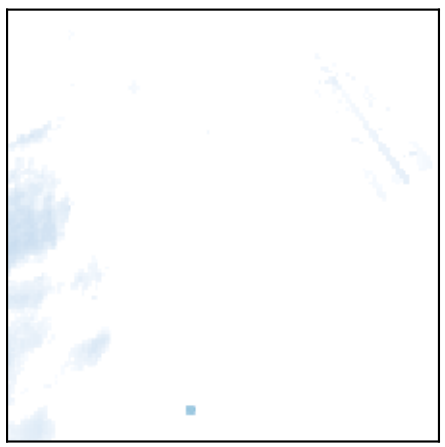


difference (mm)

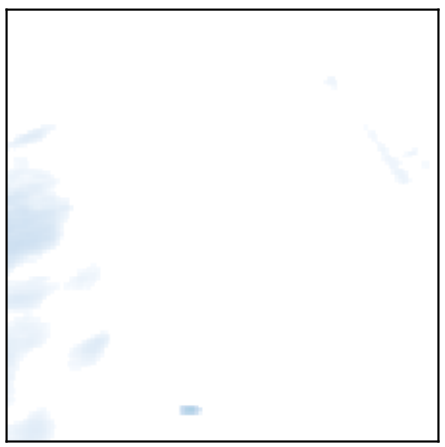


random
2024-05-03 02:00
69.6N 22.4E

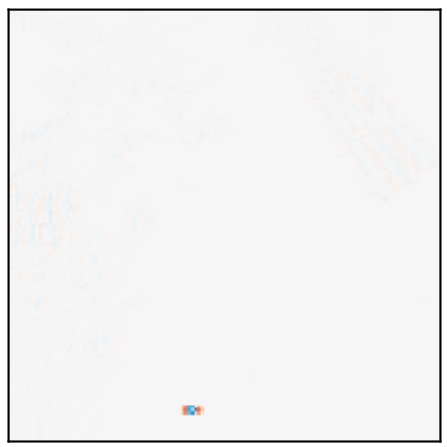
plain sum (max 1.5 mm)



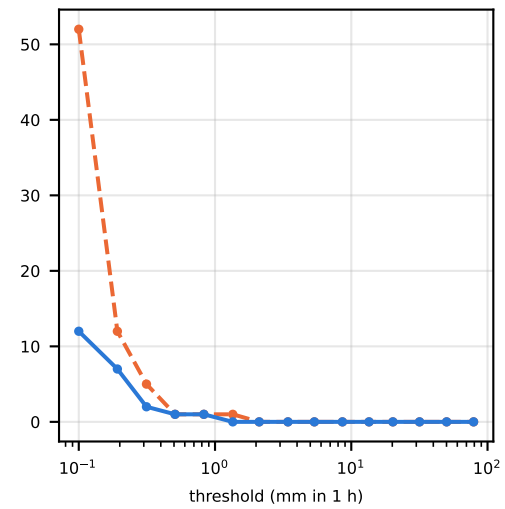
advection-corrected (max 0.9 mm)



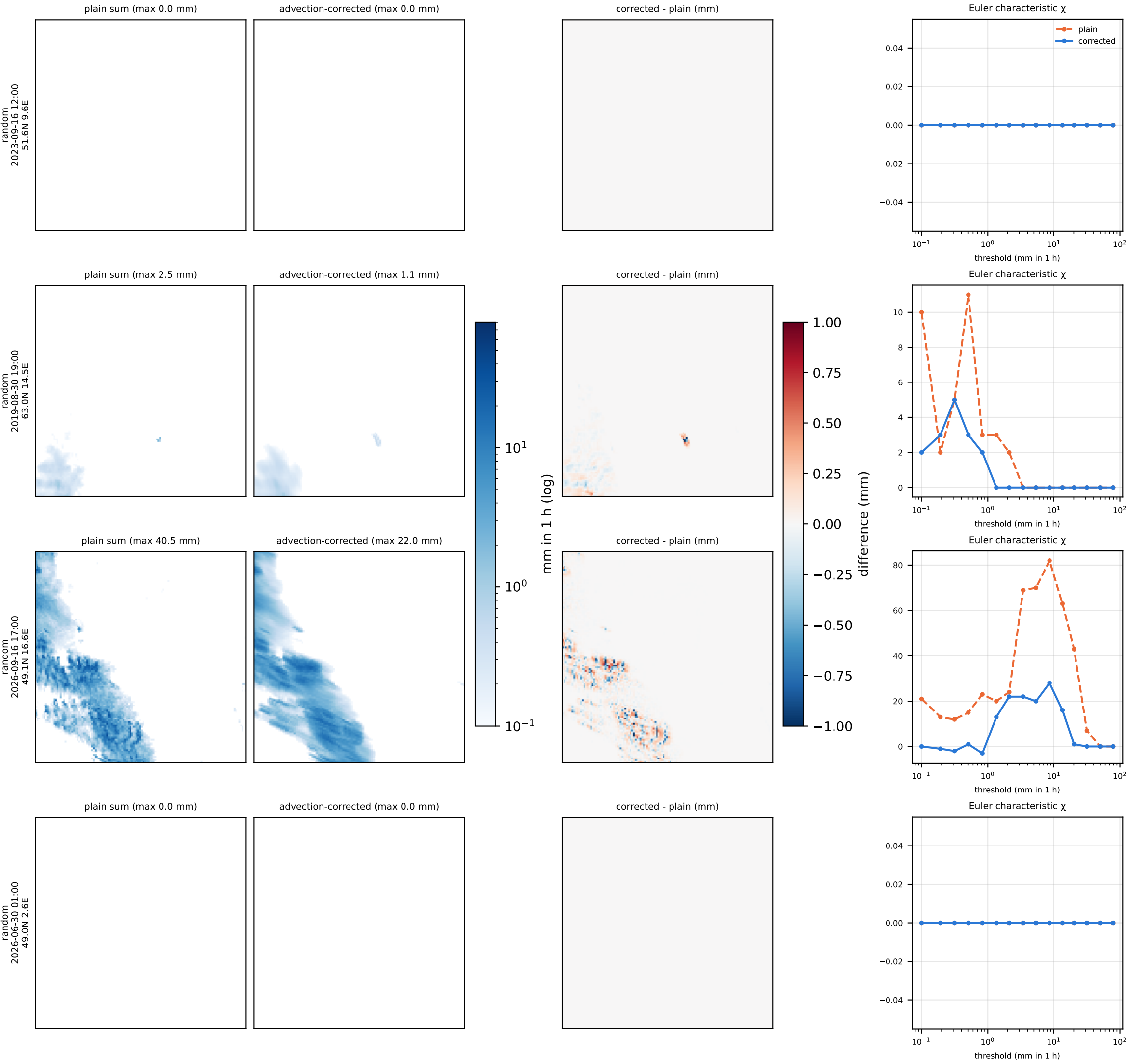
corrected - plain (mm)



Euler characteristic χ



Plain vs advection-corrected — samples 25-28 of 100



Plain vs advection-corrected — samples 29-32 of 100

plain sum (max 3.9 mm)

advection-corrected (max 1.9 mm)

corrected - plain (mm)

Euler characteristic χ

random
2018-06-17 12:00
62.4N 24.4E

plain

corrected

threshold (mm in 1 h)

plain sum (max 6.2 mm)

advection-corrected (max 6.2 mm)

corrected - plain (mm)

Euler characteristic χ

random
2024-05-19 05:00
51.6N 9.6E

threshold (mm in 1 h)

plain sum (max 10.3 mm)

advection-corrected (max 10.3 mm)

corrected - plain (mm)

Euler characteristic χ

random
2018-07-14 22:00
46.2N 22.9E

mm in 1 h (log)

difference (mm)

threshold (mm in 1 h)

plain sum (max 1.2 mm)

advection-corrected (max 1.0 mm)

corrected - plain (mm)

Euler characteristic χ

random
2019-10-02 03:00
53.9N 9.5E

threshold (mm in 1 h)

Plain vs advection-corrected — samples 33-36 of 100

plain sum (max 0.4 mm)

advection-corrected (max 0.3 mm)

corrected - plain (mm)

Euler characteristic χ

random
2015-07-02 12:00
36.9N -1.7E

plain sum (max 0.1 mm)

advection-corrected (max 0.1 mm)

corrected - plain (mm)

Euler characteristic χ

random
2022-02-06 03:00
69.1N 28.6E

plain sum (max 2.5 mm)

advection-corrected (max 0.7 mm)

corrected - plain (mm)

Euler characteristic χ

random
2022-02-11 21:00
46.5N -0.4E

plain sum (max 0.7 mm)

advection-corrected (max 0.5 mm)

corrected - plain (mm)

Euler characteristic χ

random
2018-06-09 05:00
51.1N 20.6E

mm in 1 h (log)

difference (mm)

plain

corrected

threshold (mm in 1 h)

Plain vs advection-corrected — samples 37-40 of 100

plain sum (max 0.7 mm)

advection-corrected (max 0.6 mm)

corrected - plain (mm)

Euler characteristic χ

random
2021-04-23 12:00
66.9N 26.9E

plain sum (max 2.2 mm)

advection-corrected (max 1.7 mm)

corrected - plain (mm)

Euler characteristic χ

random
2025-03-10 13:00
60.5N 18.8E

mm in 1 h (log)

difference (mm)

plain sum (max 0.0 mm)

advection-corrected (max 0.0 mm)

corrected - plain (mm)

Euler characteristic χ

random
2025-11-10 01:00
49.2N 6.1E

plain sum (max 1.3 mm)

advection-corrected (max 0.7 mm)

corrected - plain (mm)

Euler characteristic χ

random
2019-04-18 14:00
39.6N 0.8E

plain

corrected

threshold (mm in 1 h)

Plain vs advection-corrected — samples 41-44 of 100

plain sum (max 25.9 mm)
advection-corrected (max 22.6 mm)
corrected - plain (mm)
Euler characteristic χ
convective
2020-08-03 06:00
51.5N 5.9E
plain
corrected
threshold (mm in 1 h)

plain sum (max 31.0 mm)
advection-corrected (max 25.9 mm)
corrected - plain (mm)
Euler characteristic χ
convective
2018-07-14 10:00
69.1N 28.6E
threshold (mm in 1 h)

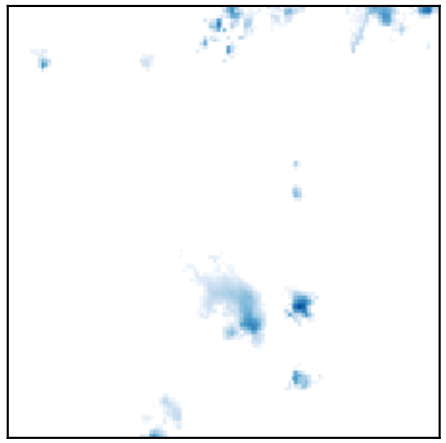
plain sum (max 29.4 mm)
advection-corrected (max 22.0 mm)
corrected - plain (mm)
Euler characteristic χ
convective
2015-07-03 07:00
51.6N 9.6E
threshold (mm in 1 h)

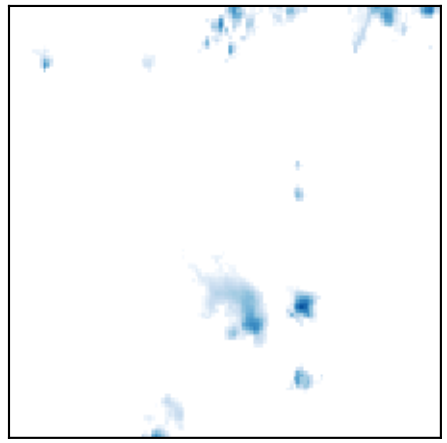
plain sum (max 21.7 mm)
advection-corrected (max 21.7 mm)
corrected - plain (mm)
Euler characteristic χ
convective
2025-08-01 19:00
56.1N 13.6E
threshold (mm in 1 h)

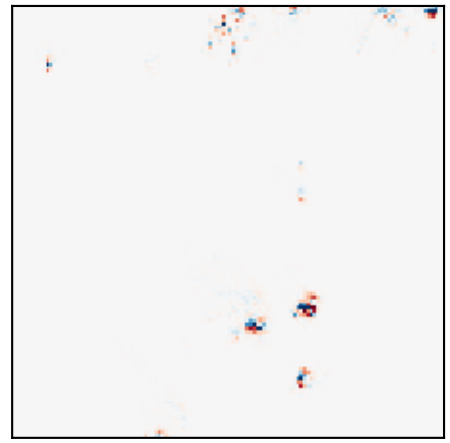
mm in 1 h (log)
difference (mm)

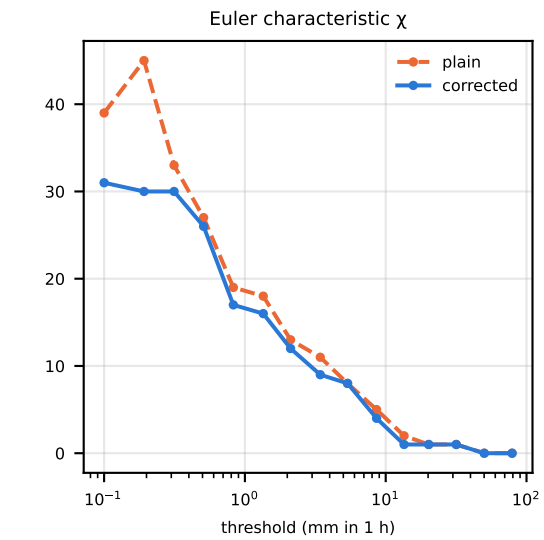
Plain vs advection-corrected — samples 45-48 of 100

convective
2016-07-08 16:00
41.5N -2.6E

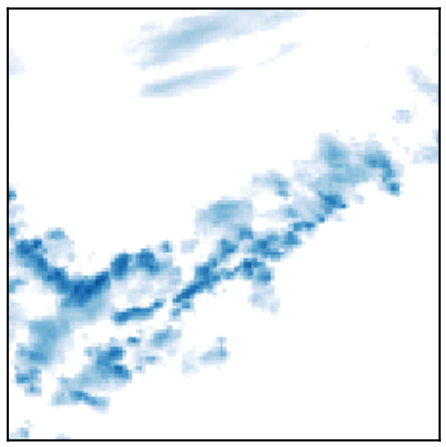


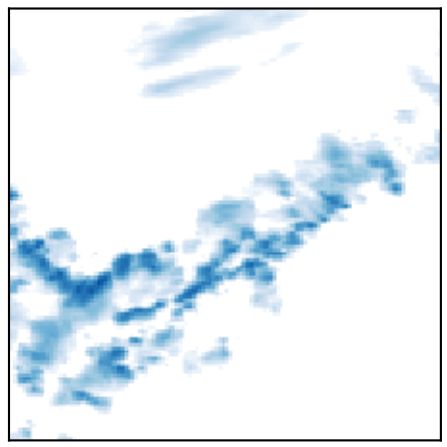


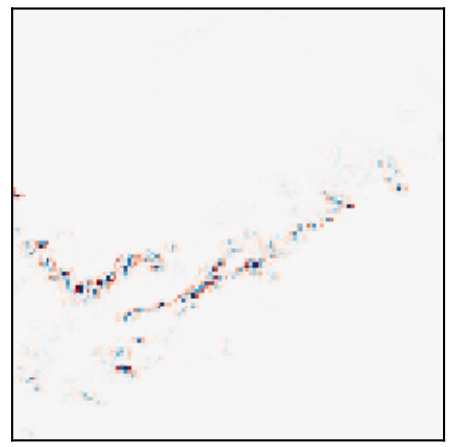


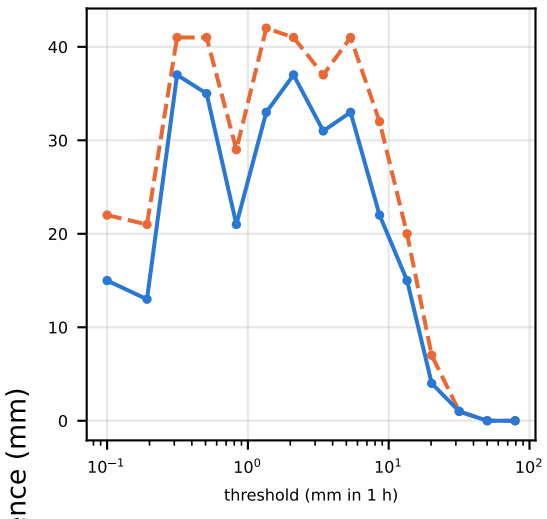


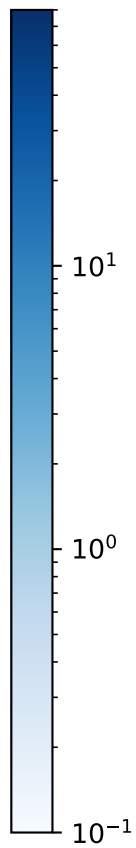
convective
2021-06-06 14:00
49.2N 13.1E

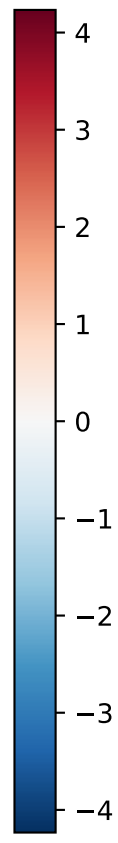




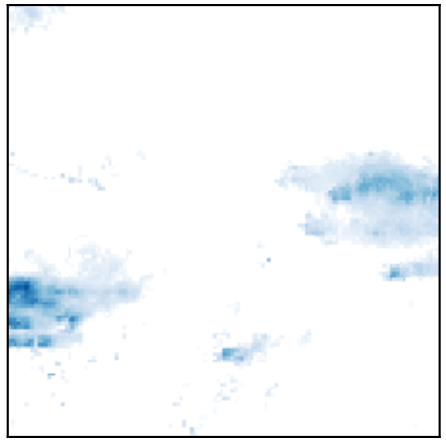




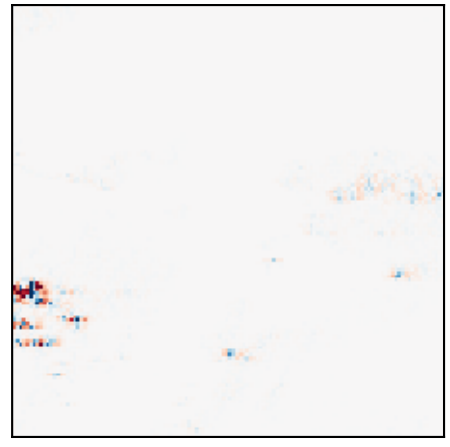


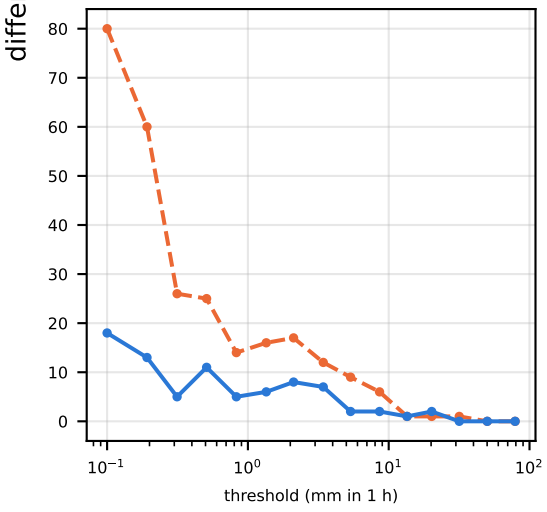


convective
2023-07-19 19:00
45.7N 26.2E

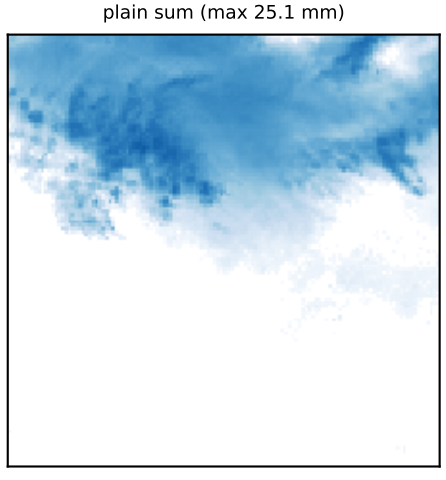


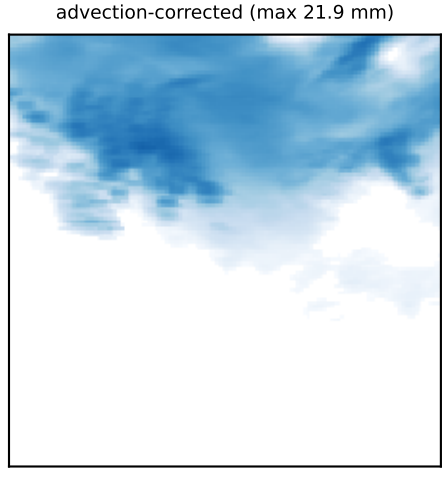


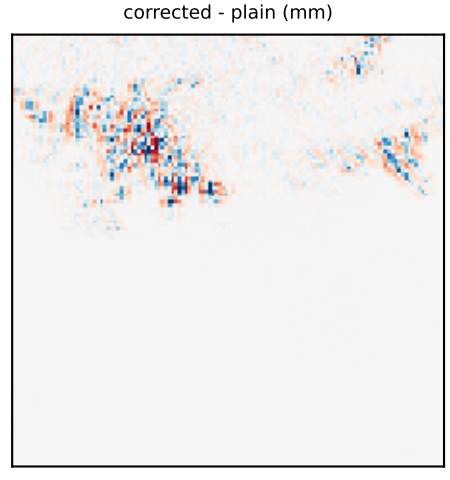


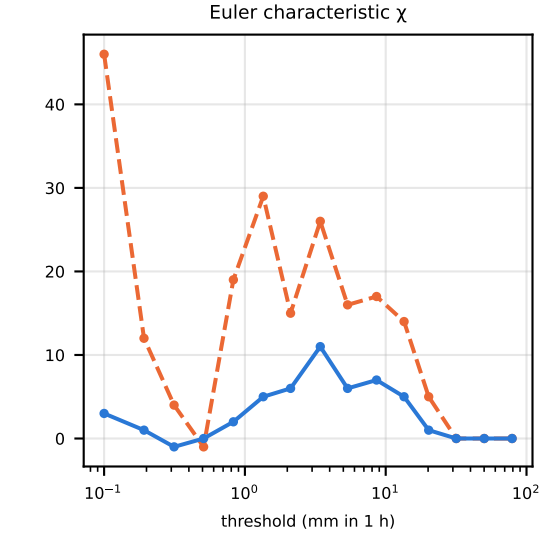


convective
2021-07-17 23:00
46.5N 19.6E





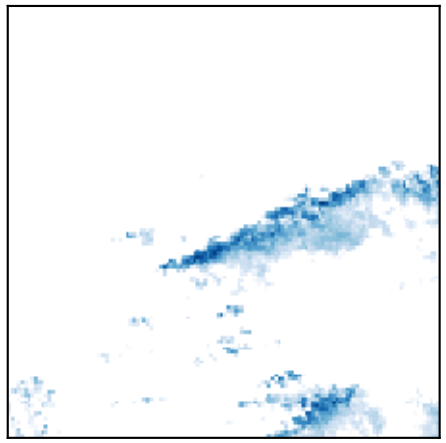




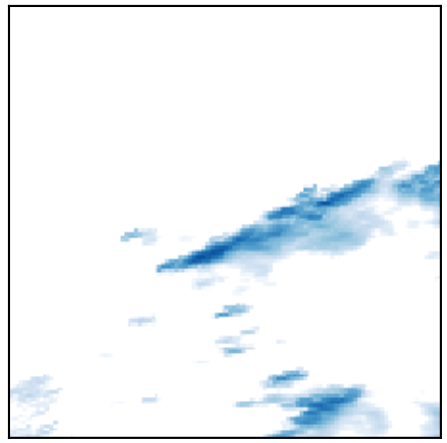
Plain vs advection-corrected — samples 49-52 of 100

convective
2024-08-11 14:00
60.1N 23.4E

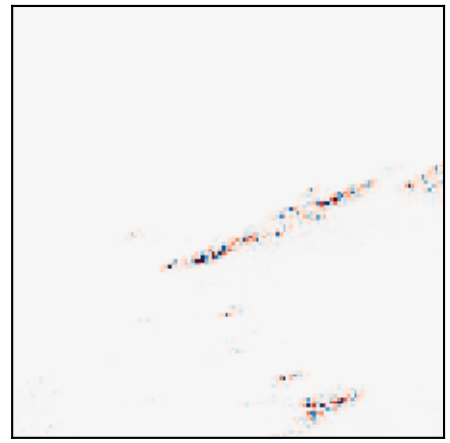
plain sum (max 57.8 mm)



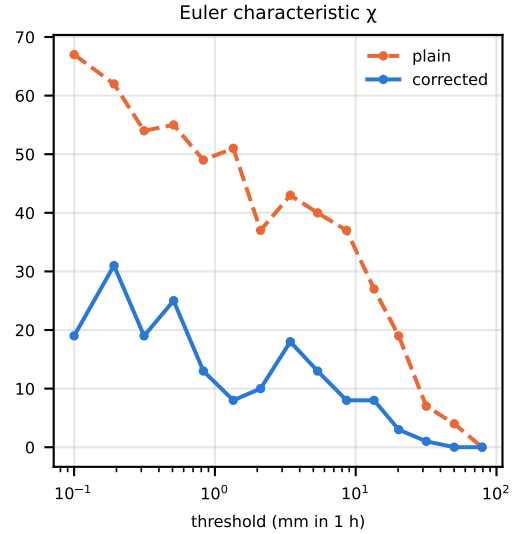
advection-corrected (max 32.8 mm)



corrected - plain (mm)

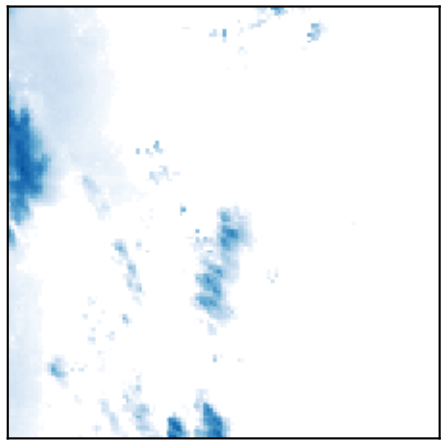


Euler characteristic χ

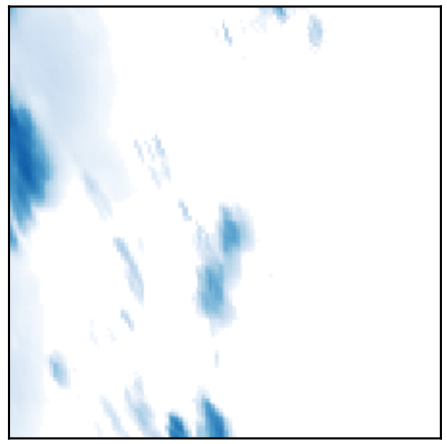


convective
2021-06-19 18:00
49.2N 6.1E

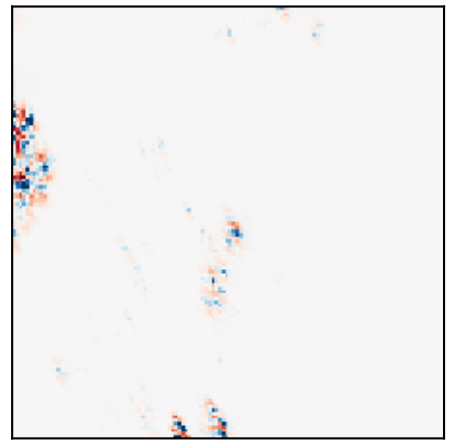
plain sum (max 32.6 mm)



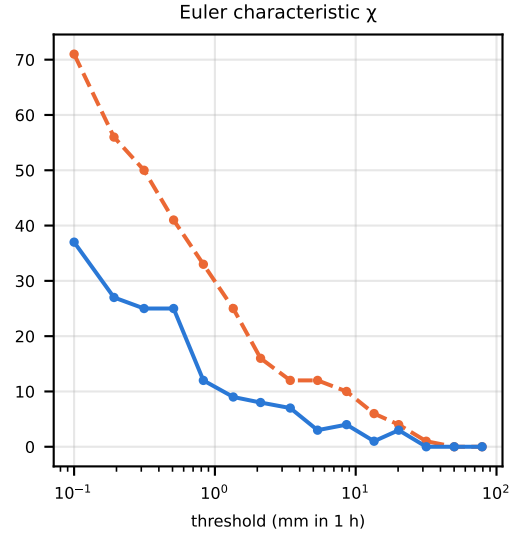
advection-corrected (max 24.9 mm)

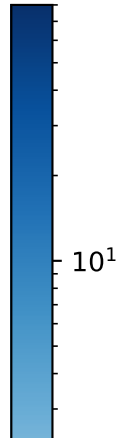


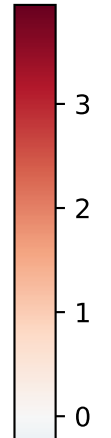
corrected - plain (mm)



Euler characteristic χ

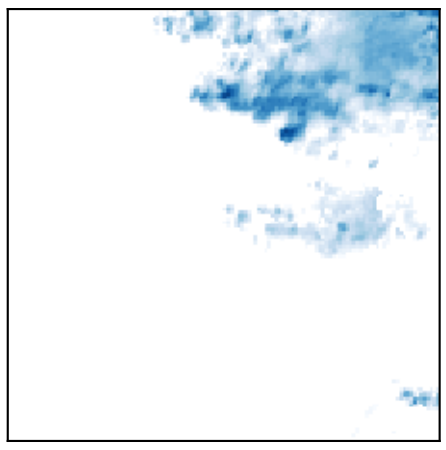




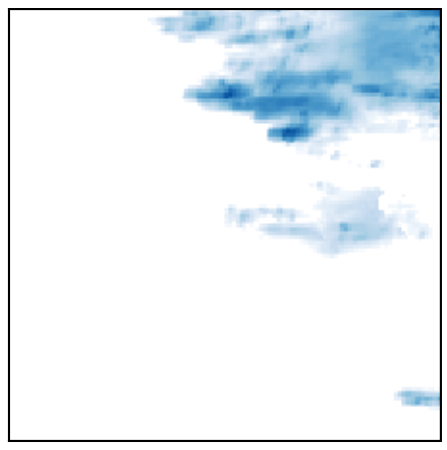


convective
2016-06-29 19:00
41.5N -2.6E

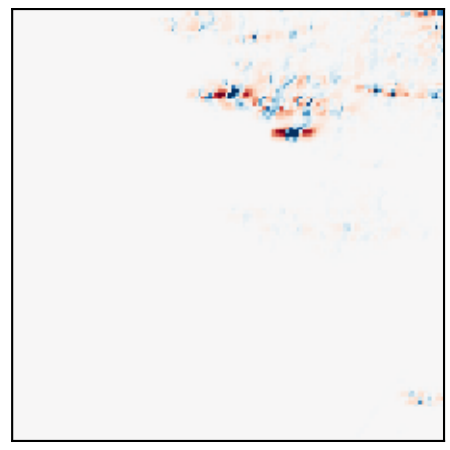
plain sum (max 49.6 mm)



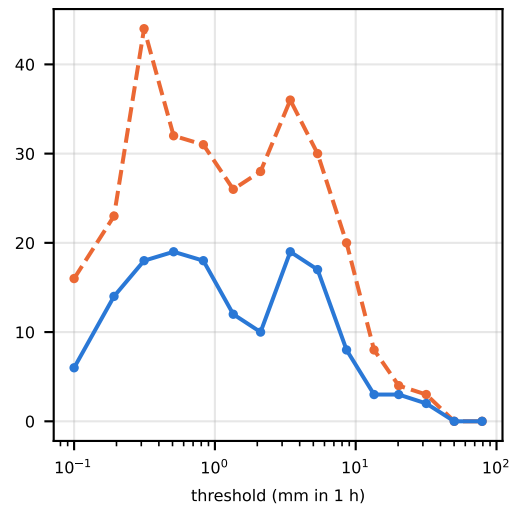
advection-corrected (max 32.1 mm)



corrected - plain (mm)

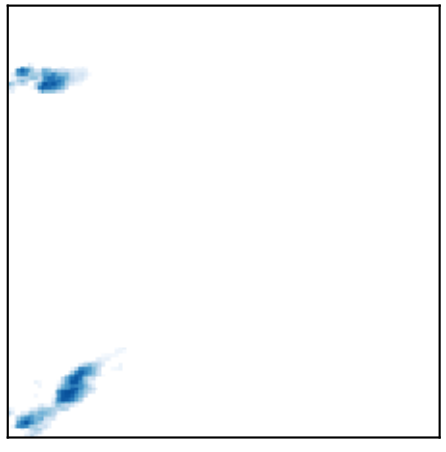


Euler characteristic χ

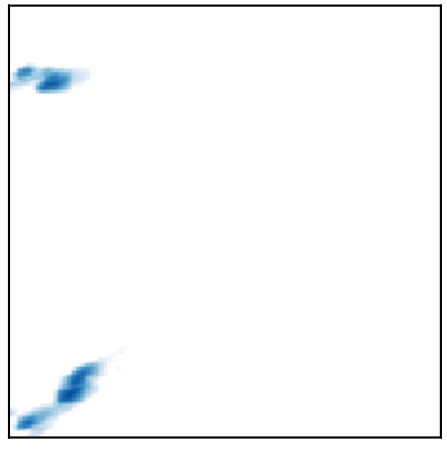


convective
2021-08-28 16:00
39.6N 0.8E

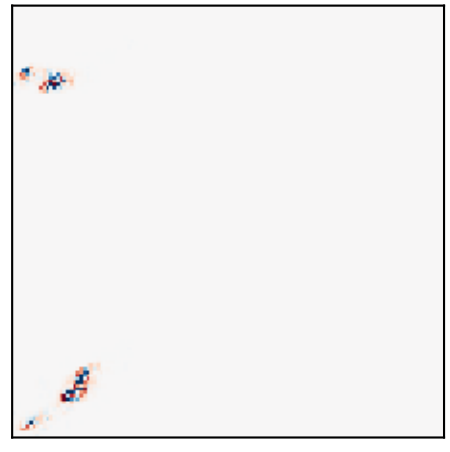
plain sum (max 37.5 mm)



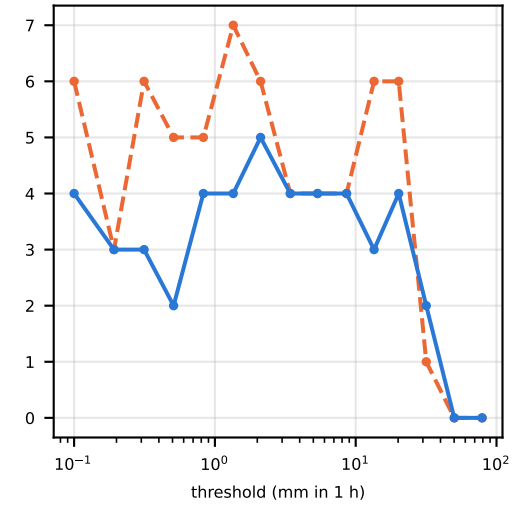
advection-corrected (max 34.7 mm)



corrected - plain (mm)



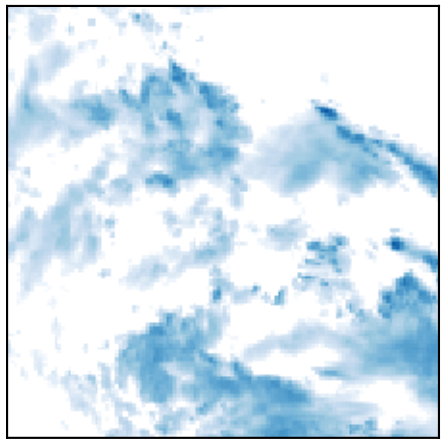
Euler characteristic χ



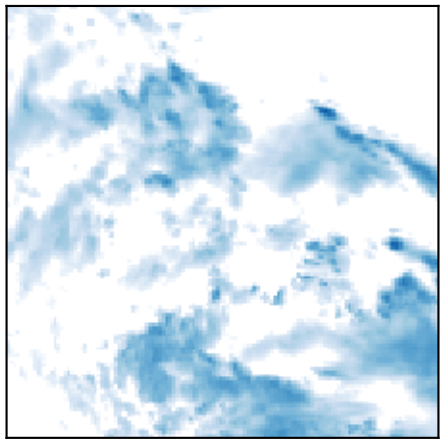
Plain vs advection-corrected — samples 53-56 of 100

convective
2019-08-12 16:00
60.8N 9.5E

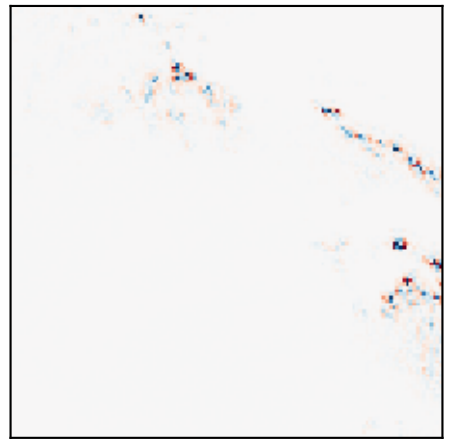
plain sum (max 27.8 mm)



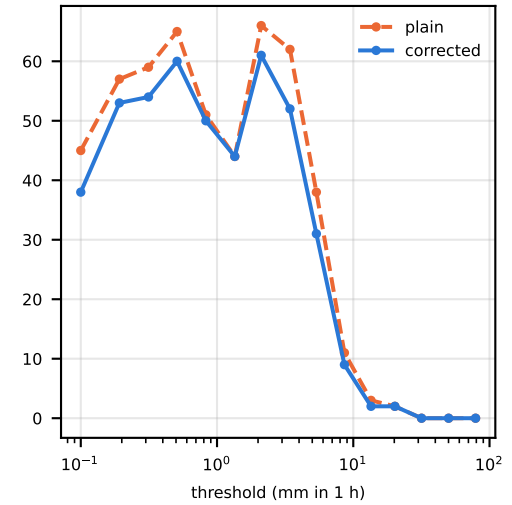
advection-corrected (max 25.3 mm)



corrected - plain (mm)

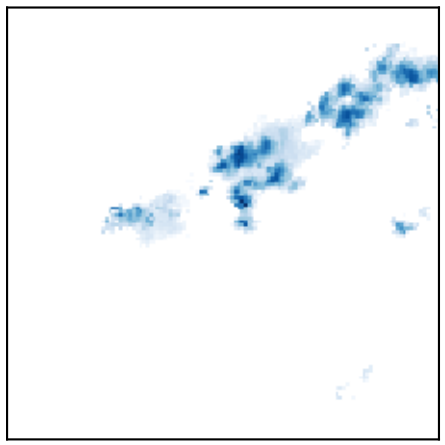


Euler characteristic χ

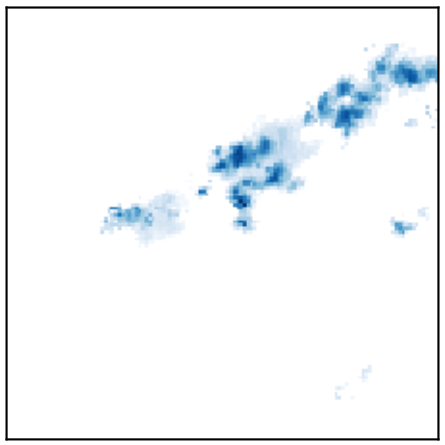


convective
2022-08-26 15:00
44.5N 16.0E

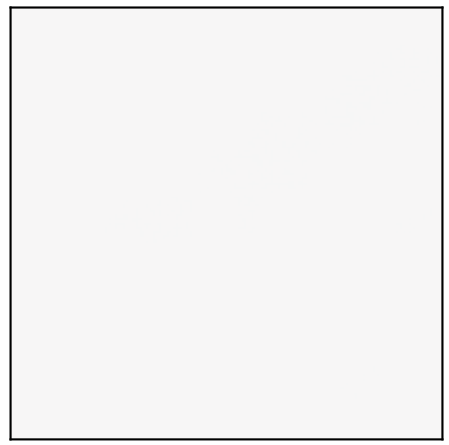
plain sum (max 79.6 mm)



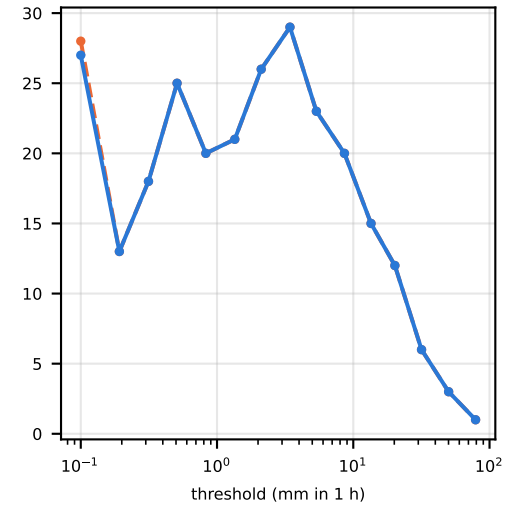
advection-corrected (max 79.6 mm)

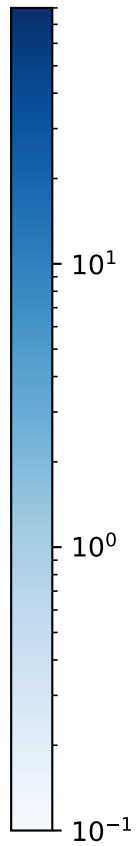


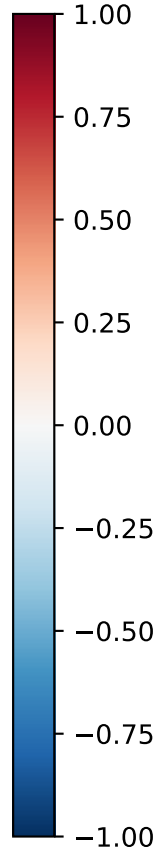
corrected - plain (mm)



Euler characteristic χ

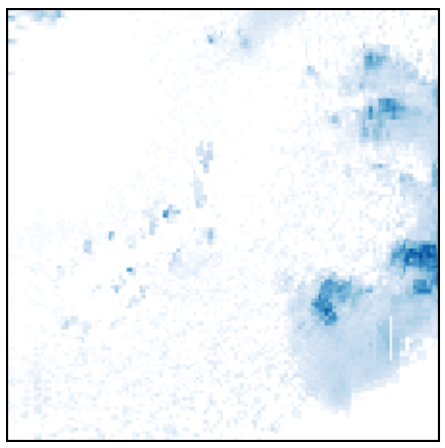




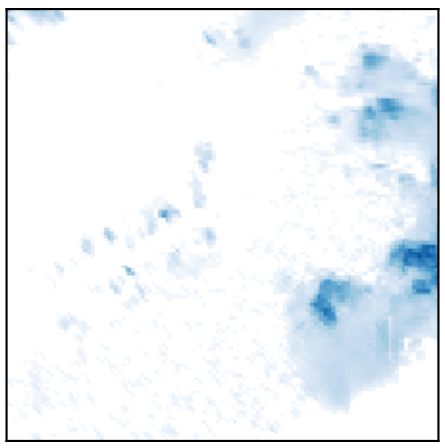


convective
2015-06-15 14:00
48.4N 23.5E

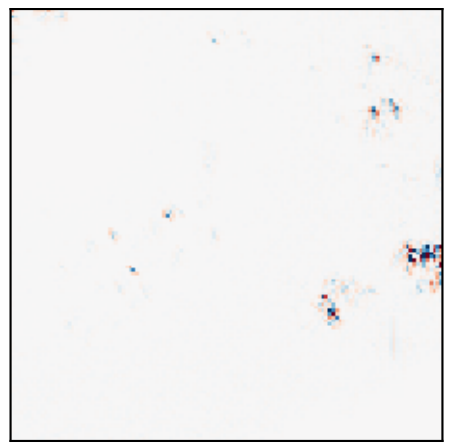
plain sum (max 35.6 mm)



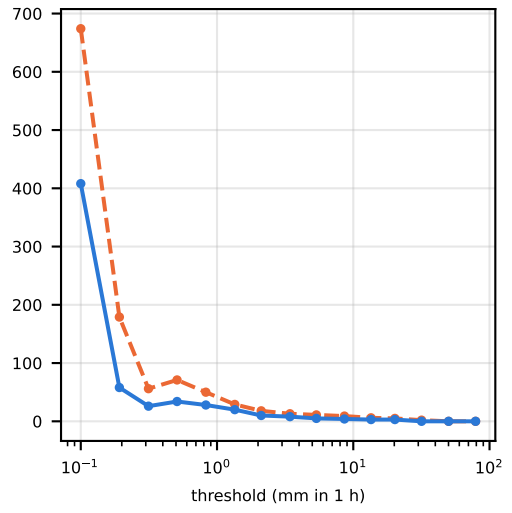
advection-corrected (max 29.1 mm)



corrected - plain (mm)

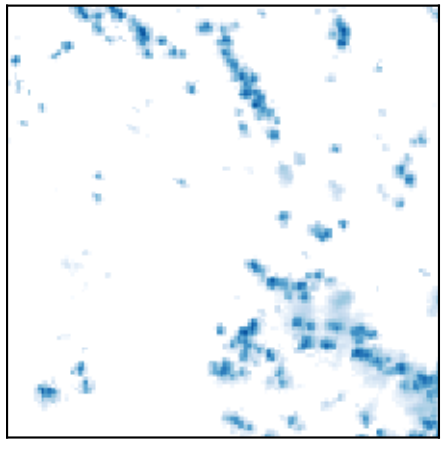


Euler characteristic χ

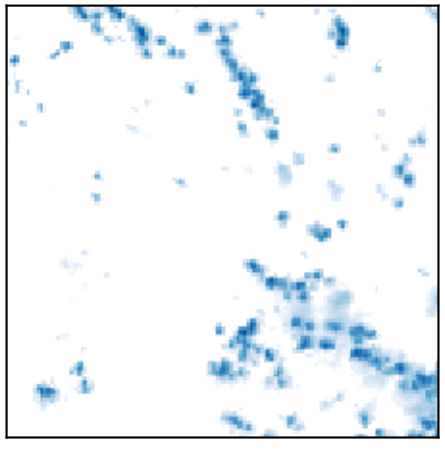


convective
2023-06-28 09:00
62.4N 24.4E

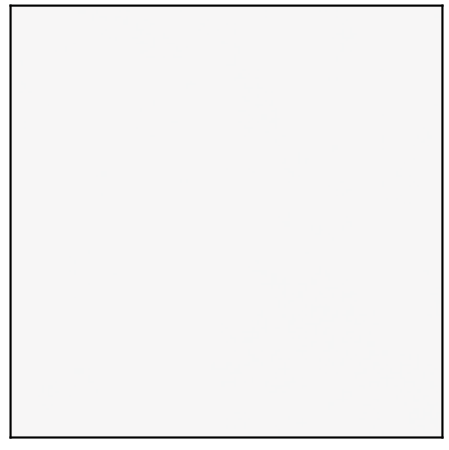
plain sum (max 31.1 mm)



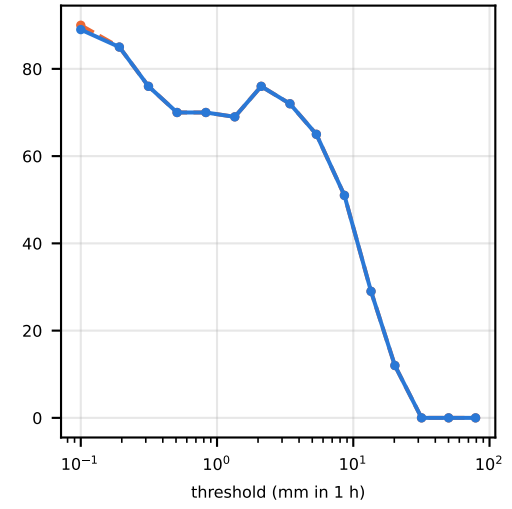
advection-corrected (max 31.1 mm)



corrected - plain (mm)

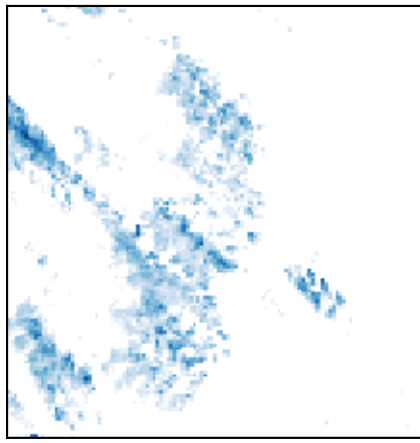


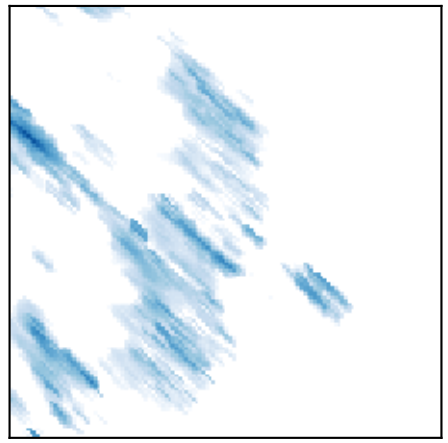
Euler characteristic χ

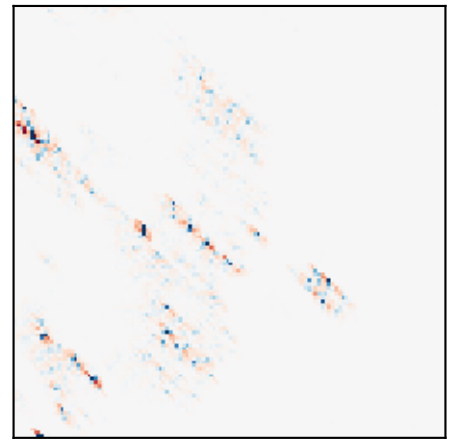


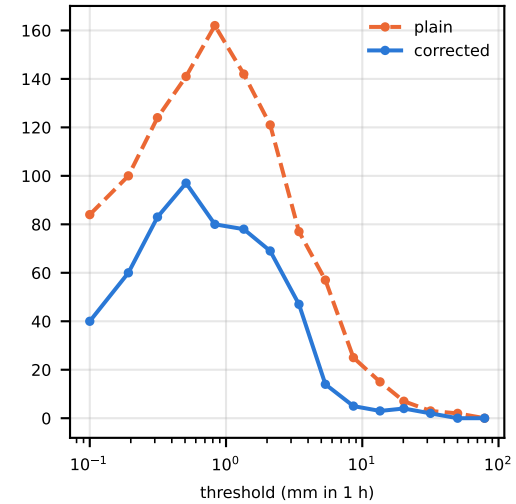
Plain vs advection-corrected — samples 57-60 of 100

convective
2024-07-16 14:00
51.6N 9.6E

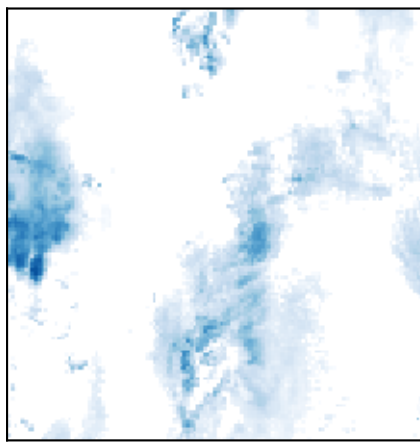


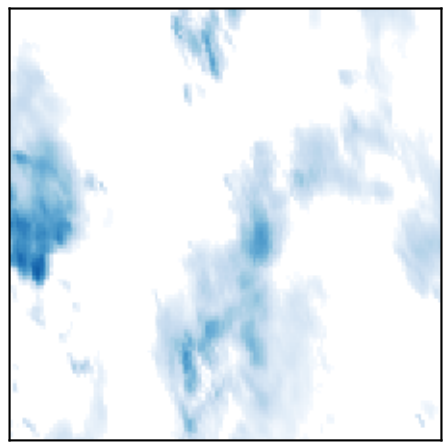


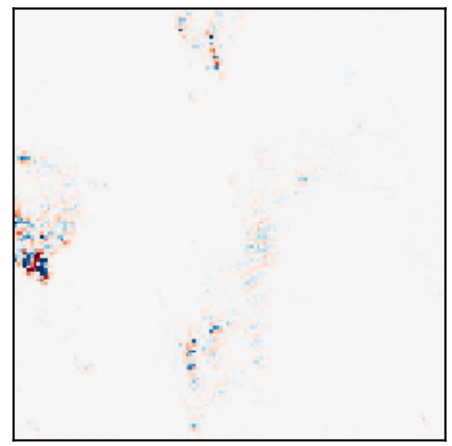


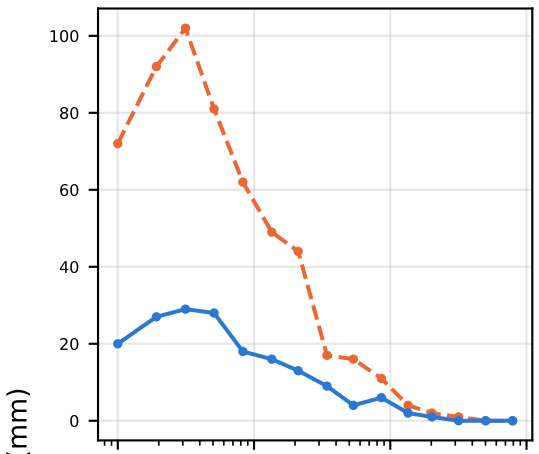


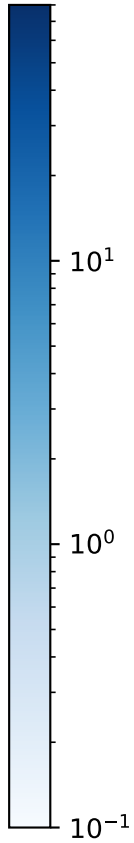
convective
2019-06-09 20:00
45.7N 26.2E

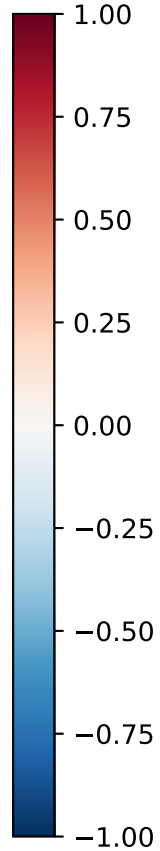




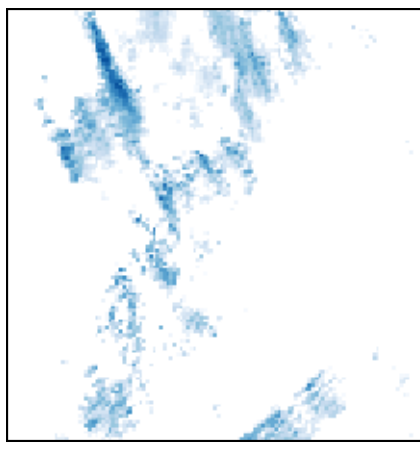


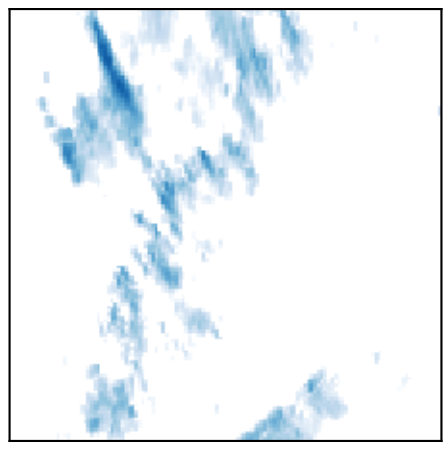


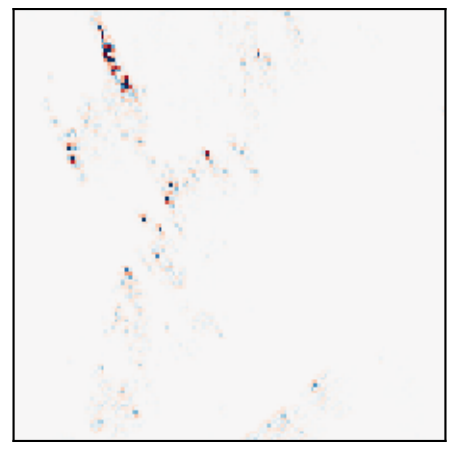


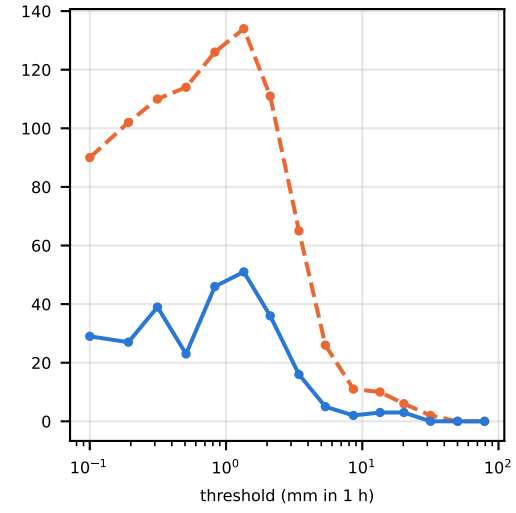


convective
2025-08-19 05:00
46.7N 2.9E

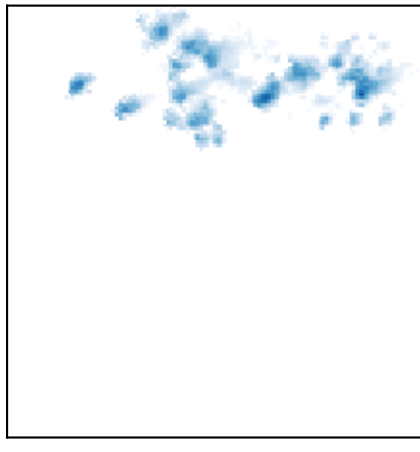


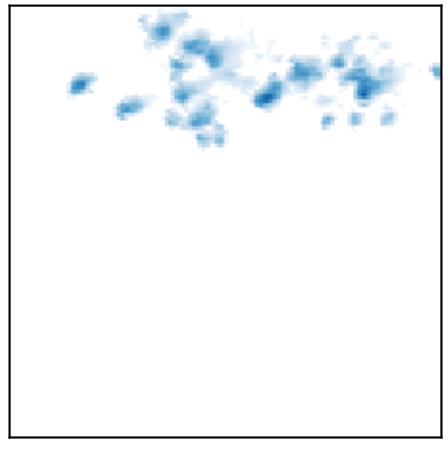


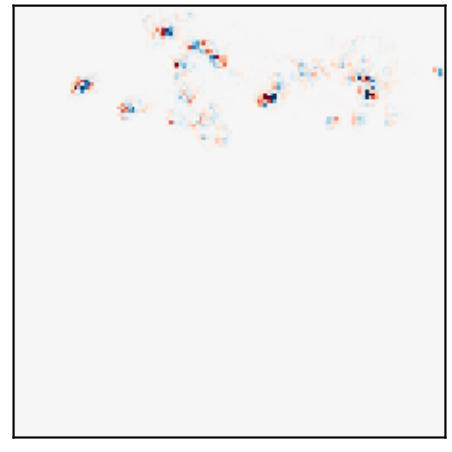


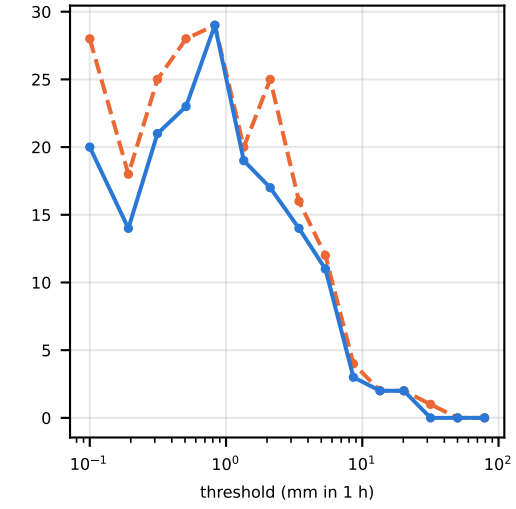


convective
2020-07-22 15:00
43.3N -6.2E





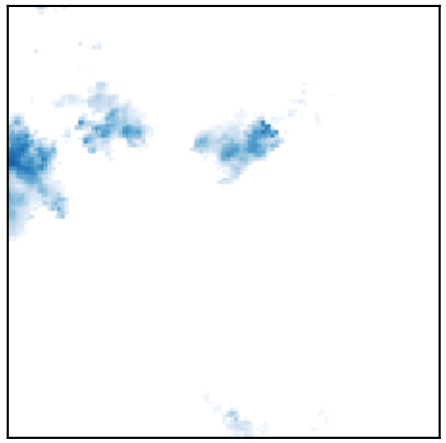




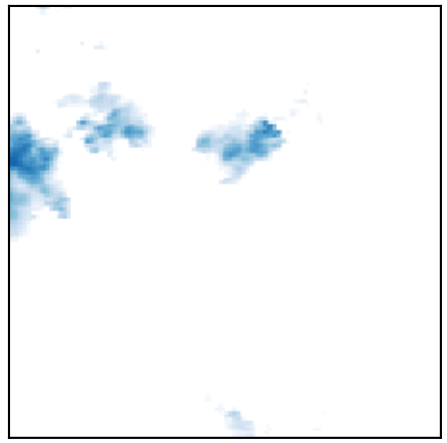
Plain vs advection-corrected — samples 61-64 of 100

convective
2019-07-30 20:00
46.8N 16.3E

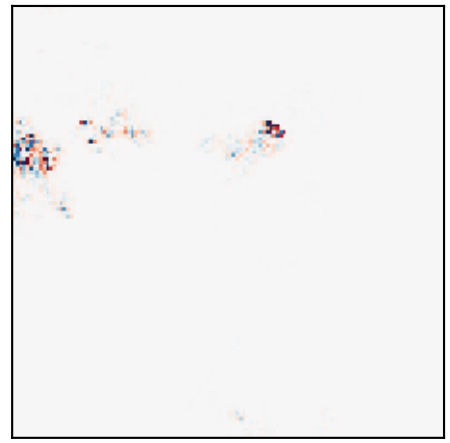
plain sum (max 22.3 mm)



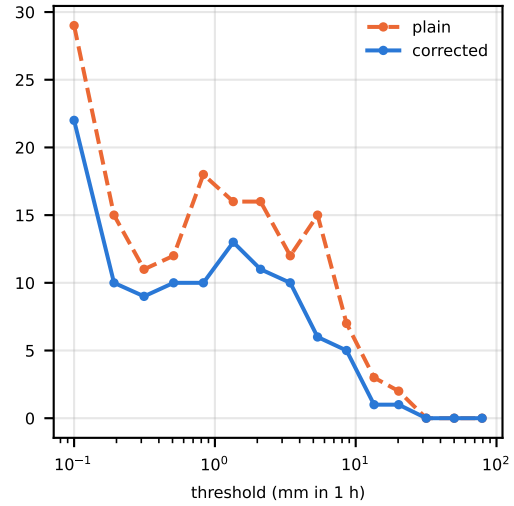
advection-corrected (max 20.7 mm)



corrected - plain (mm)

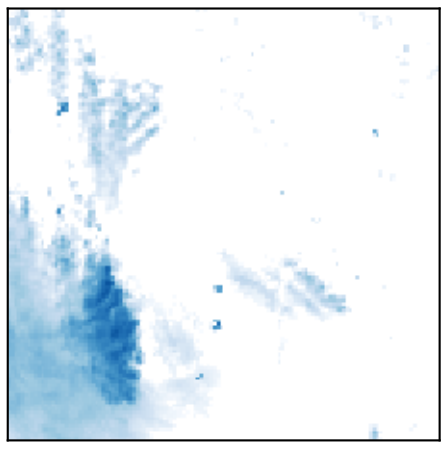


Euler characteristic χ

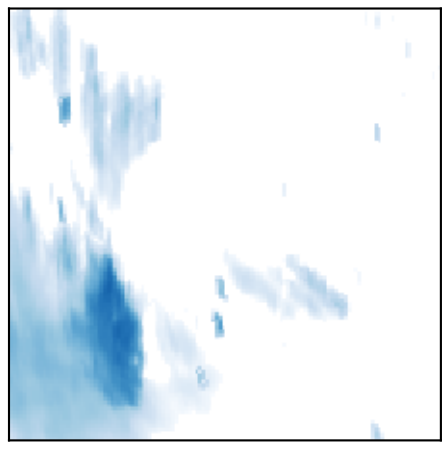


convective
2022-08-19 00:00
62.4N 24.4E

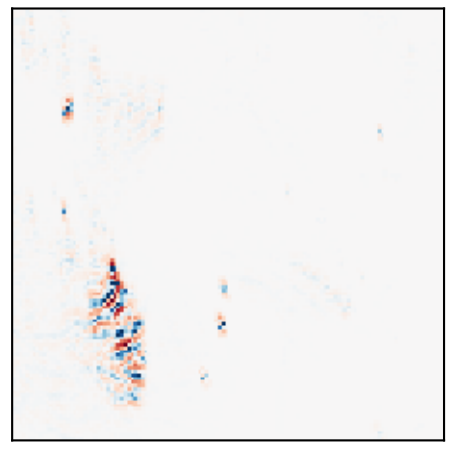
plain sum (max 31.3 mm)



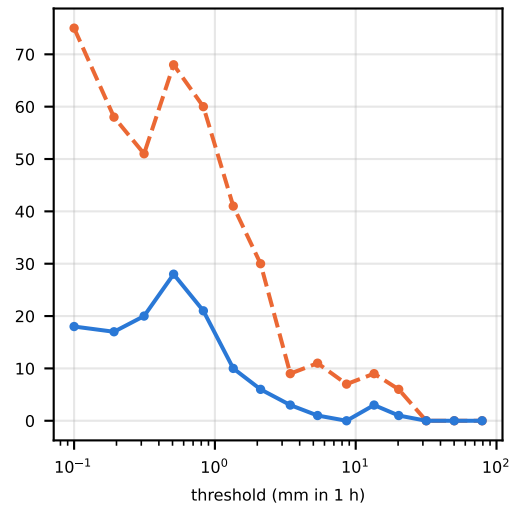
advection-corrected (max 20.8 mm)



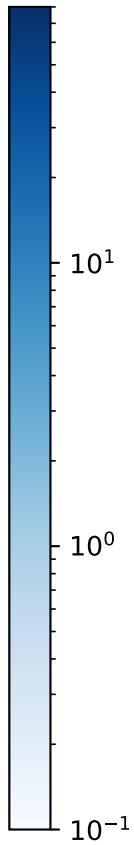
corrected - plain (mm)



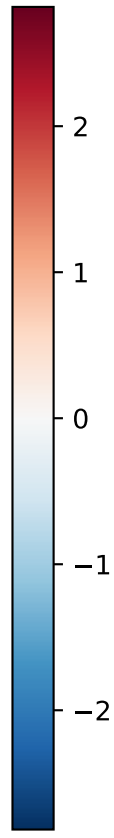
Euler characteristic χ



mm in 1 h (log)

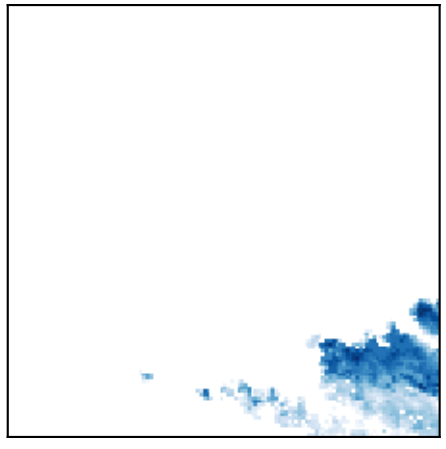


difference (mm)

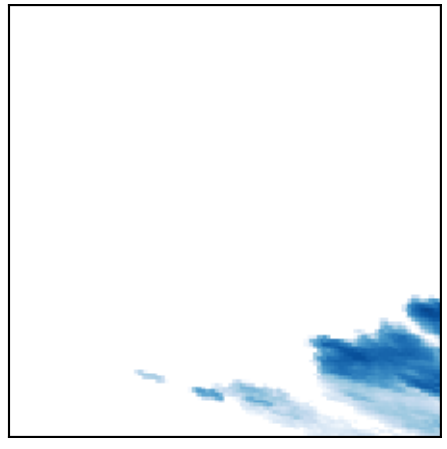


convective
2024-07-11 15:00
44.4N 3.2E

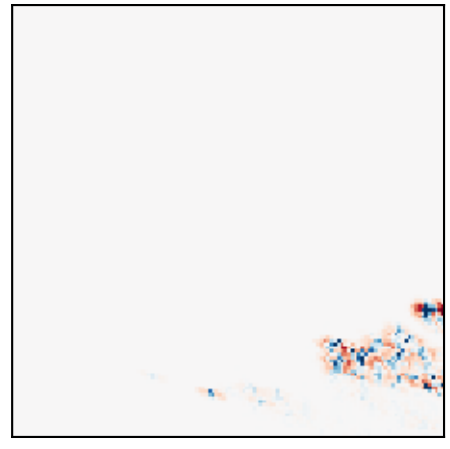
plain sum (max 91.2 mm)



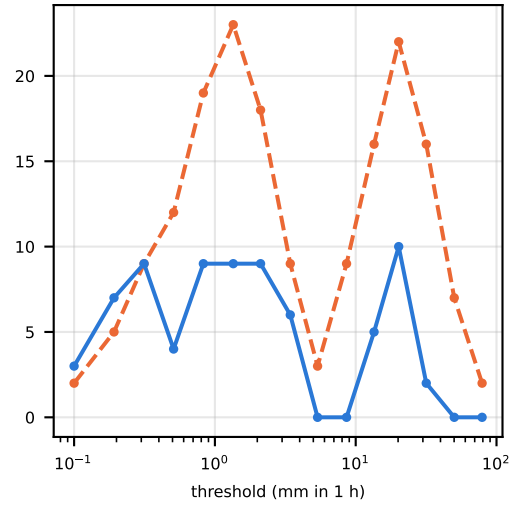
advection-corrected (max 44.0 mm)



corrected - plain (mm)

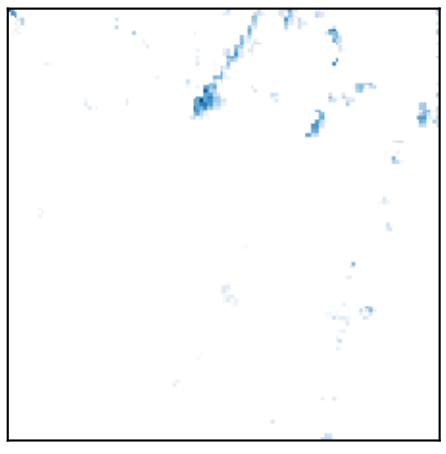


Euler characteristic χ

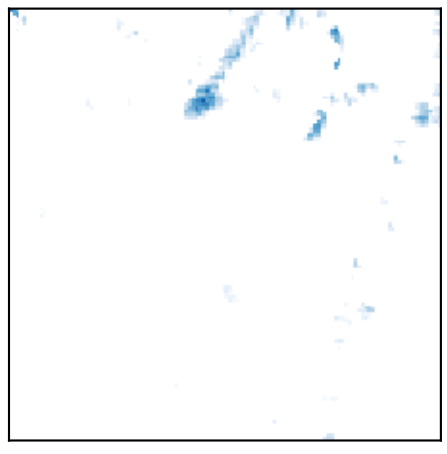


convective
2018-07-13 04:00
58.5N 9.5E

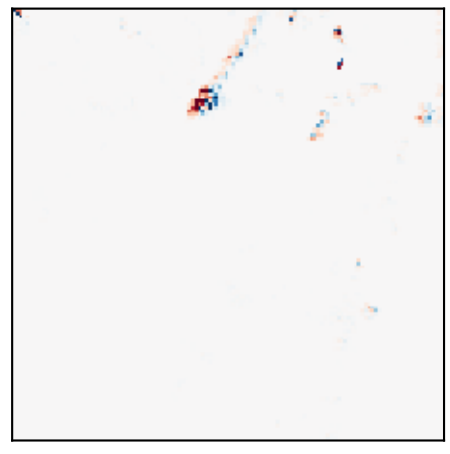
plain sum (max 36.9 mm)



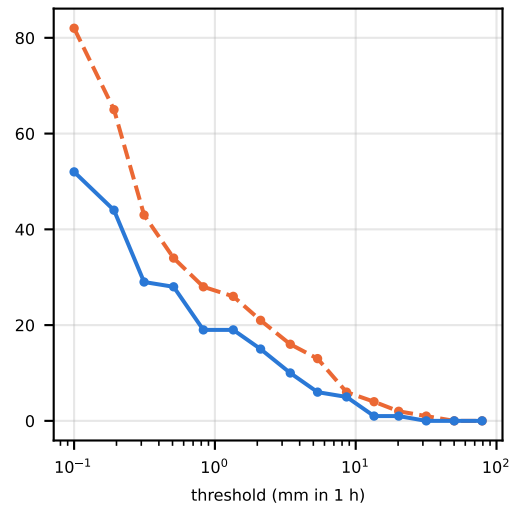
advection-corrected (max 24.6 mm)



corrected - plain (mm)



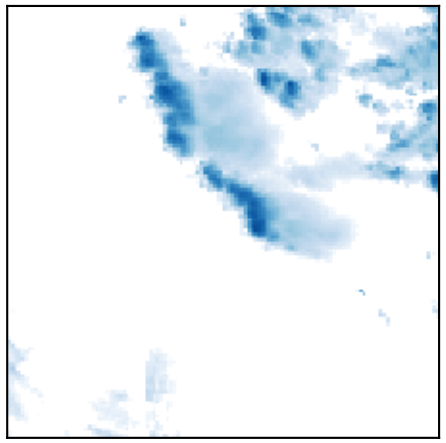
Euler characteristic χ



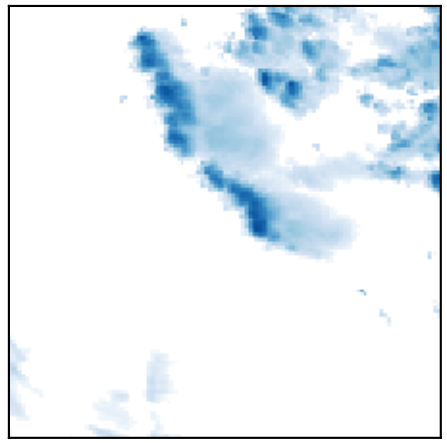
Plain vs advection-corrected — samples 65-68 of 100

convective
2019-07-03 14:00
46.9N 6.3E

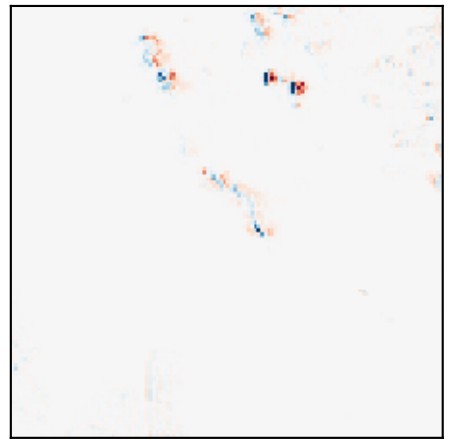
plain sum (max 38.9 mm)



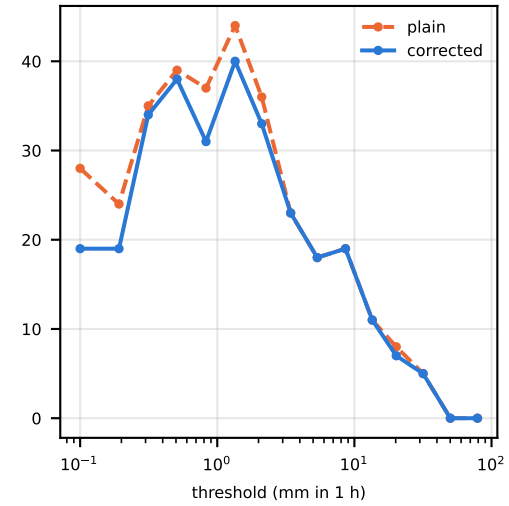
advection-corrected (max 38.2 mm)



corrected - plain (mm)

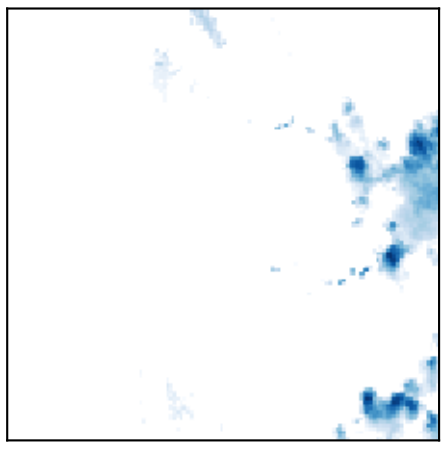


Euler characteristic χ

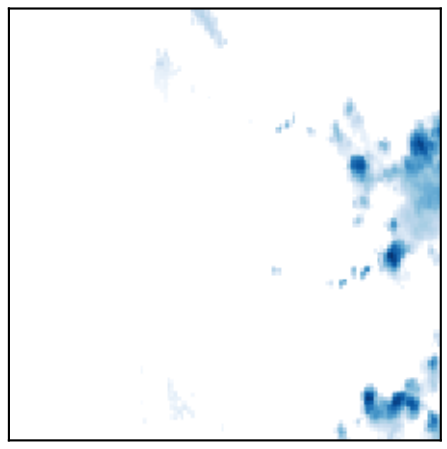


convective
2018-06-28 18:00
40.5N -8.6E

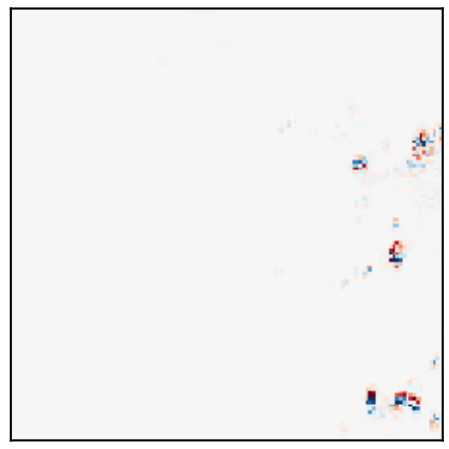
plain sum (max 96.4 mm)



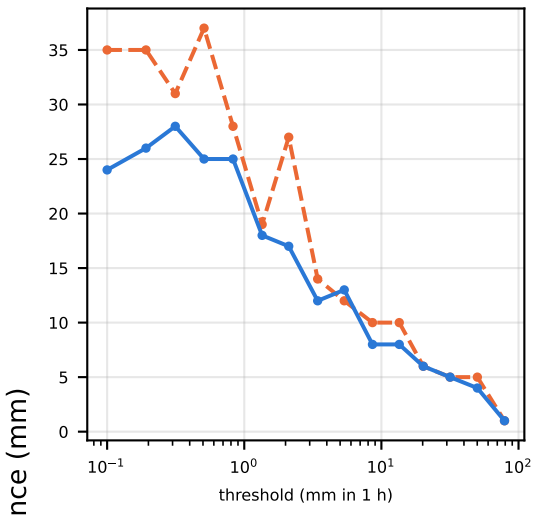
advection-corrected (max 85.6 mm)



corrected - plain (mm)

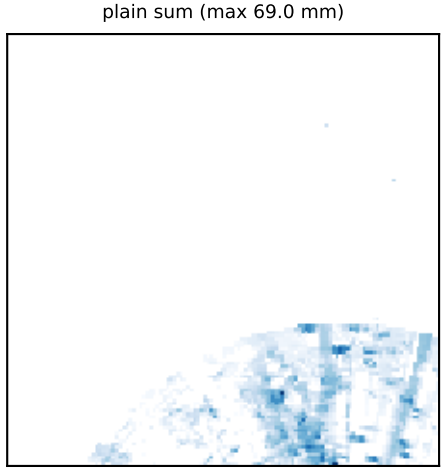


Euler characteristic χ

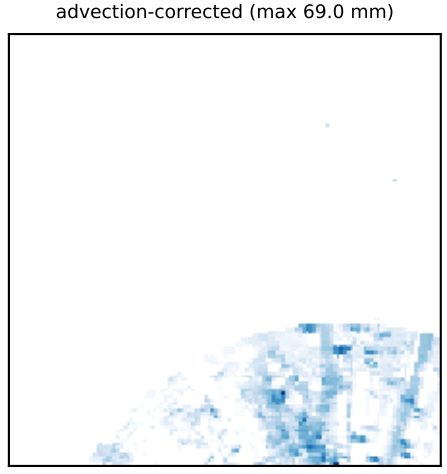


convective
2019-07-25 00:00
51.5N 5.9E

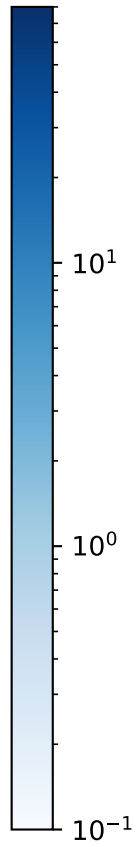
plain sum (max 69.0 mm)



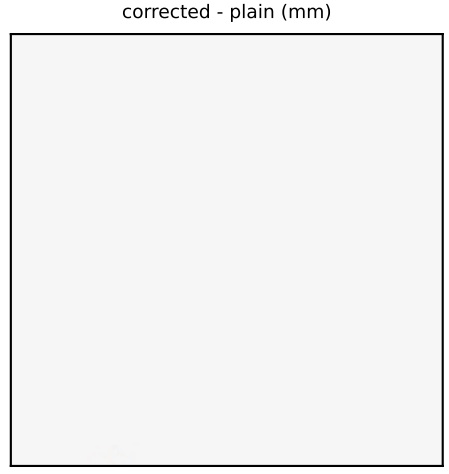
advection-corrected (max 69.0 mm)



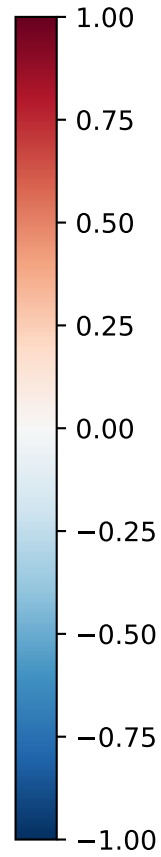
mm in 1 h (log)



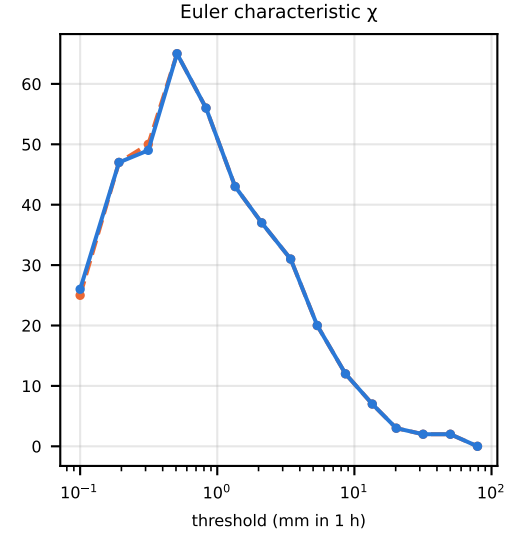
corrected - plain (mm)



difference (mm)

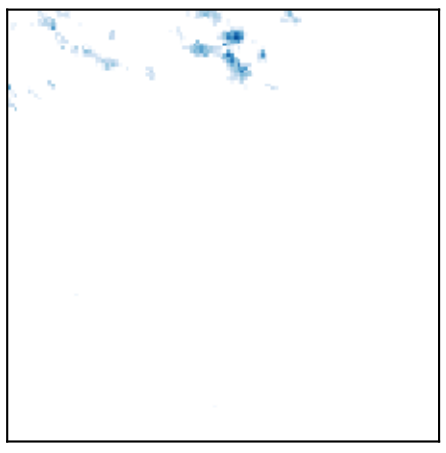


Euler characteristic χ

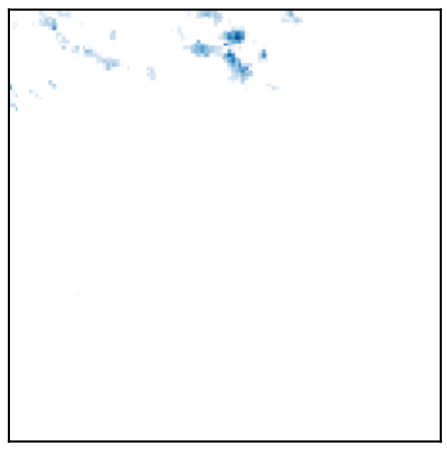


convective
2024-08-04 17:00
61.8N 29.1E

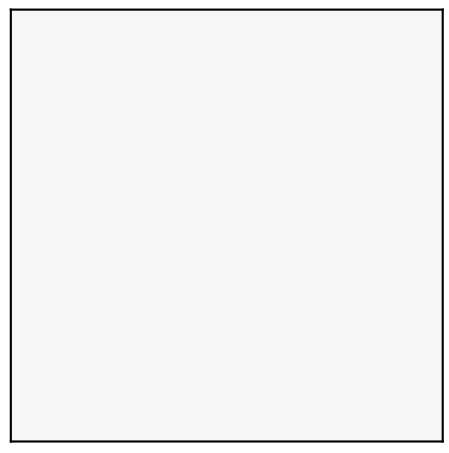
plain sum (max 37.1 mm)



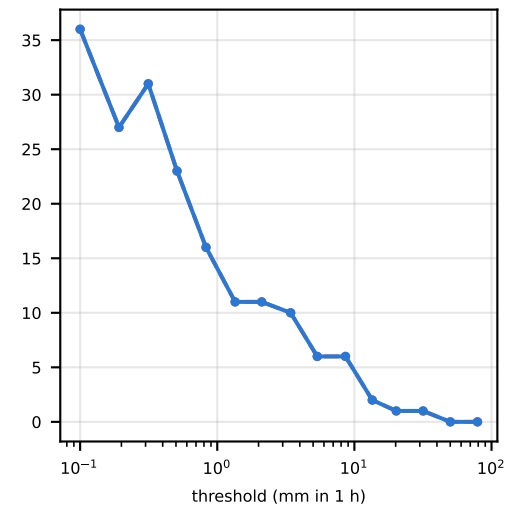
advection-corrected (max 37.1 mm)



corrected - plain (mm)



Euler characteristic χ



Plain vs advection-corrected — samples 69-72 of 100

plain sum (max 53.0 mm)
advection-corrected (max 31.6 mm)
corrected - plain (mm)
Euler characteristic χ
convective
2025-07-23 19:00
46.5N -0.4E
plain
corrected
threshold (mm in 1 h)

plain sum (max 31.1 mm)
advection-corrected (max 24.4 mm)
corrected - plain (mm)
Euler characteristic χ
convective
2015-07-30 10:00
39.6N 0.8E
threshold (mm in 1 h)

plain sum (max 89.0 mm)
advection-corrected (max 35.2 mm)
corrected - plain (mm)
Euler characteristic χ
convective
2021-07-31 23:00
46.9N 13.0E
mm in 1 h (log)
difference (mm)
threshold (mm in 1 h)

plain sum (max 31.8 mm)
advection-corrected (max 31.6 mm)
corrected - plain (mm)
Euler characteristic χ
convective
2021-06-18 18:00
46.8N 16.3E
threshold (mm in 1 h)

Plain vs advection-corrected — samples 73-76 of 100

plain sum (max 58.7 mm)

advection-corrected (max 27.8 mm)

corrected - plain (mm)

Euler characteristic χ

convective
2025-06-26 09:00
49.2N 6.1E

plain sum (max 40.8 mm)

advection-corrected (max 25.8 mm)

corrected - plain (mm)

Euler characteristic χ

convective
2024-08-02 08:00
46.8N 16.3E

plain sum (max 31.1 mm)

advection-corrected (max 25.3 mm)

corrected - plain (mm)

Euler characteristic χ

convective
2020-07-14 10:00
64.0N 30.6E

plain sum (max 29.5 mm)

advection-corrected (max 26.5 mm)

corrected - plain (mm)

Euler characteristic χ

convective
2025-08-22 13:00
60.5N 18.8E

plain
corrected

threshold (mm in 1 h)

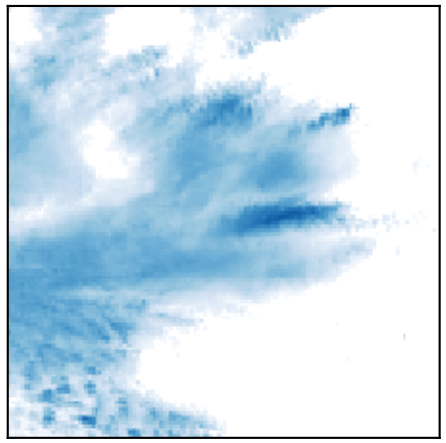
mm in 1 h (log)

difference (mm)

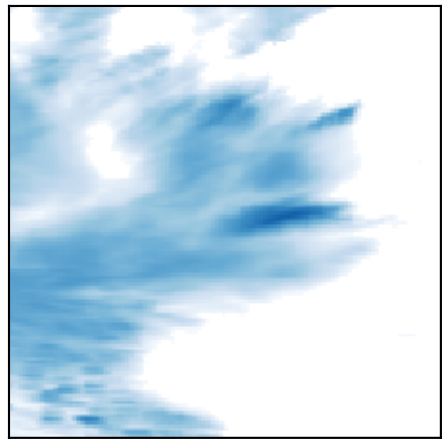
Plain vs advection-corrected — samples 77-80 of 100

convective
2020-08-07 21:00
43.9N 22.4E

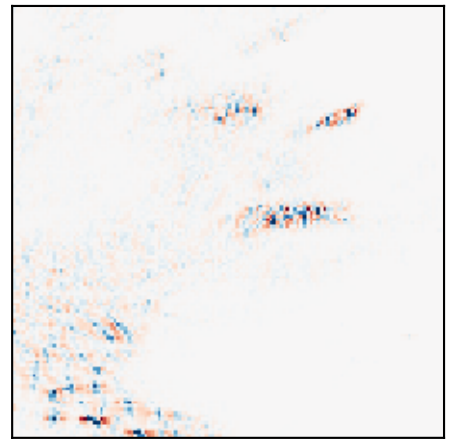
plain sum (max 26.1 mm)



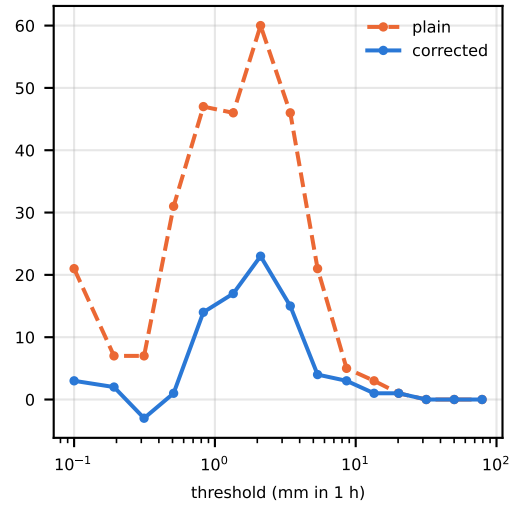
advection-corrected (max 23.4 mm)



corrected - plain (mm)

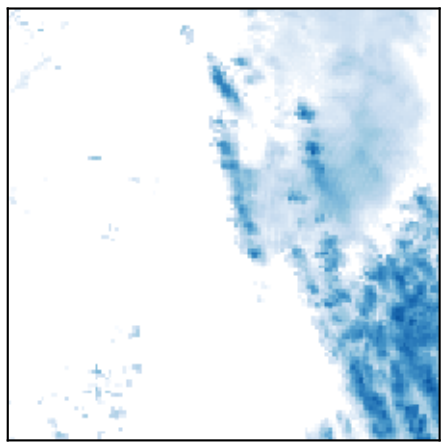


Euler characteristic χ

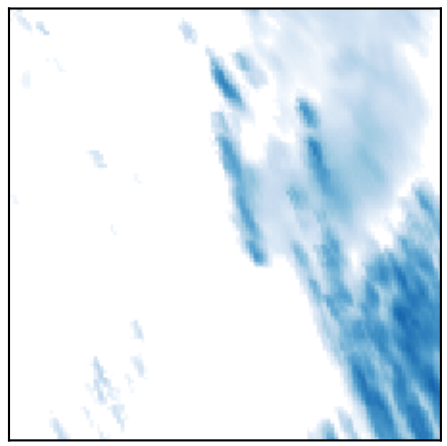


convective
2019-06-13 19:00
51.4N 16.9E

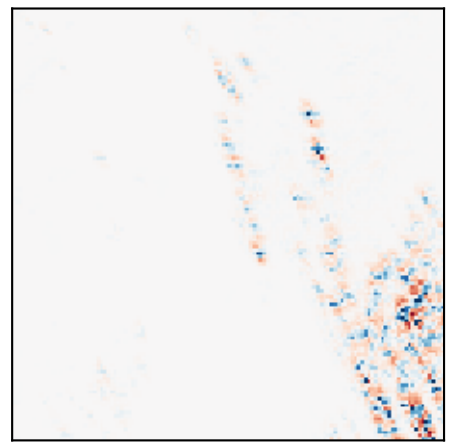
plain sum (max 31.1 mm)



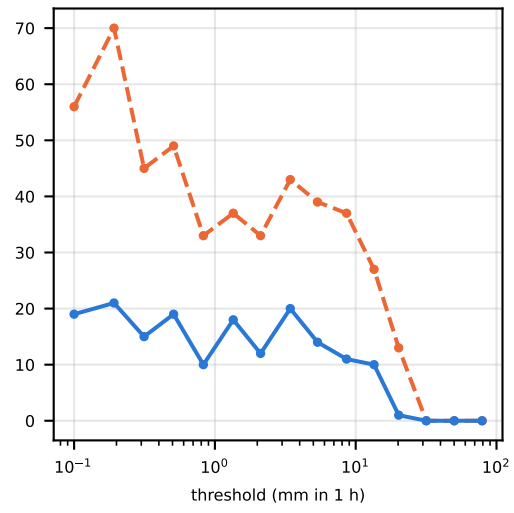
advection-corrected (max 20.2 mm)

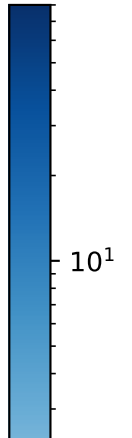


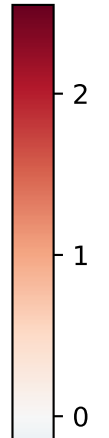
corrected - plain (mm)



Euler characteristic χ

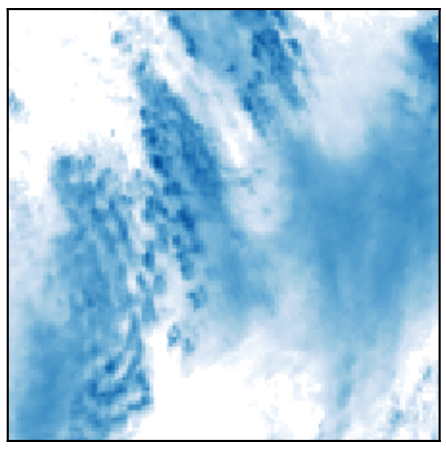




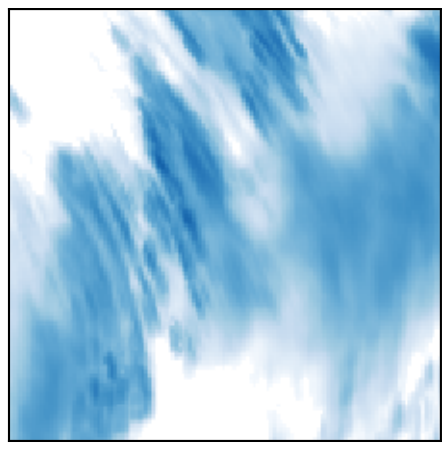


convective
2022-07-20 18:00
49.2N 6.1E

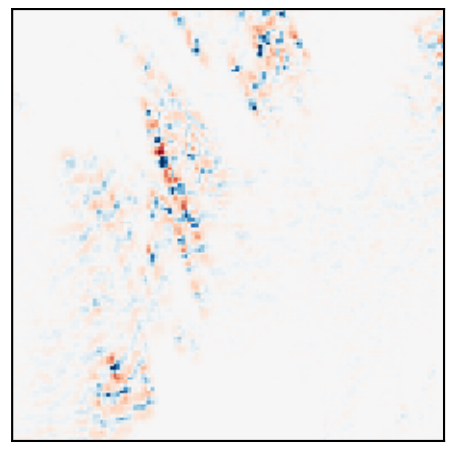
plain sum (max 46.6 mm)



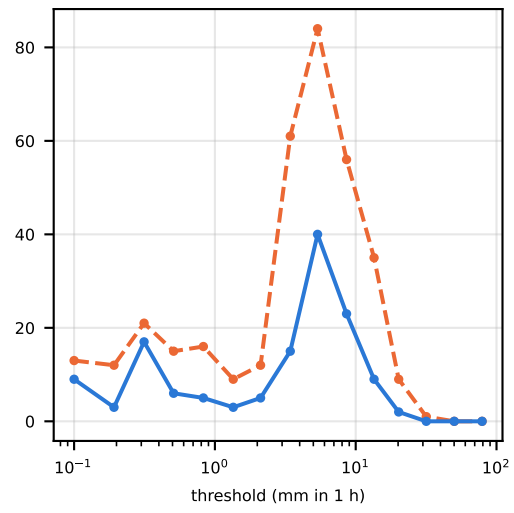
advection-corrected (max 22.5 mm)



corrected - plain (mm)

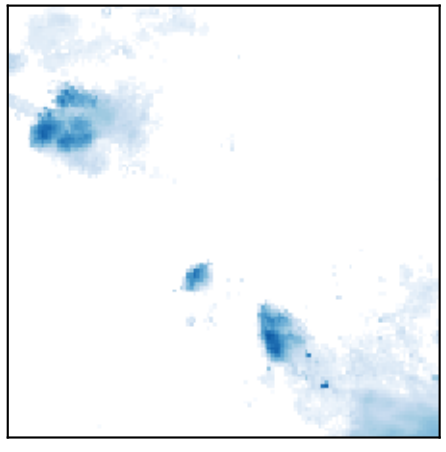


Euler characteristic χ

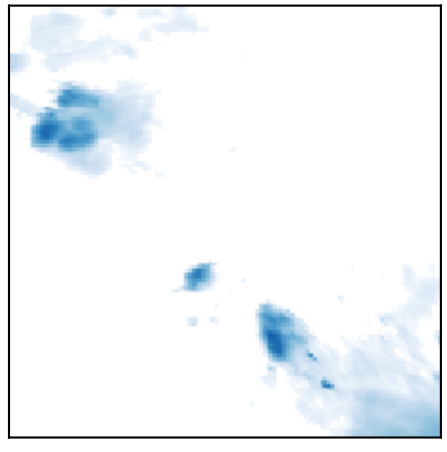


convective
2020-07-02 23:00
46.5N 19.6E

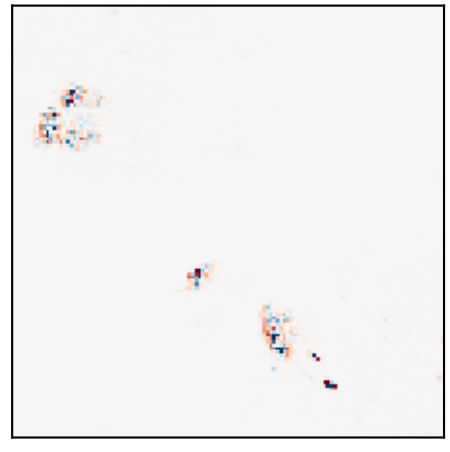
plain sum (max 27.0 mm)



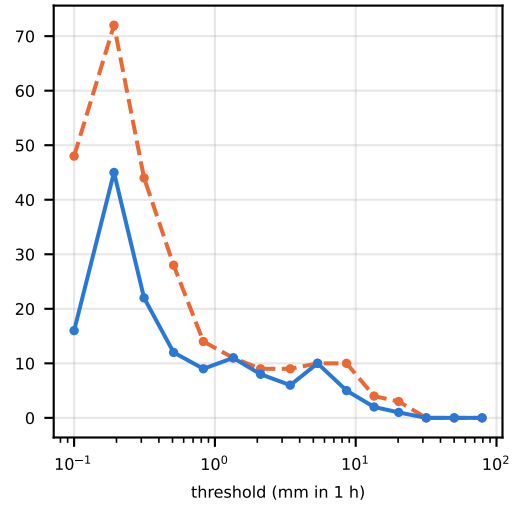
advection-corrected (max 24.0 mm)



corrected - plain (mm)



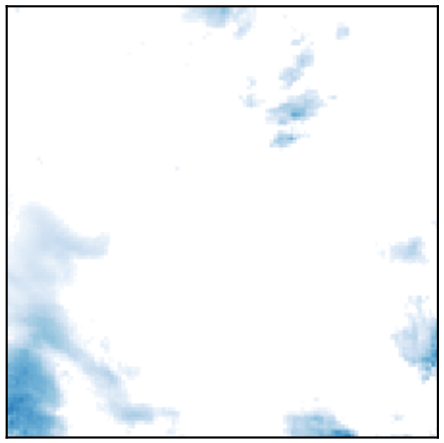
Euler characteristic χ



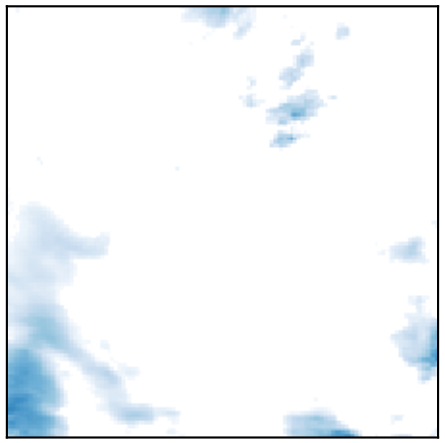
Plain vs advection-corrected — samples 81-84 of 100

event
2016-06-01 19:00
49.2N 13.1E
elvira_friederike_201605

plain sum (max 9.6 mm)



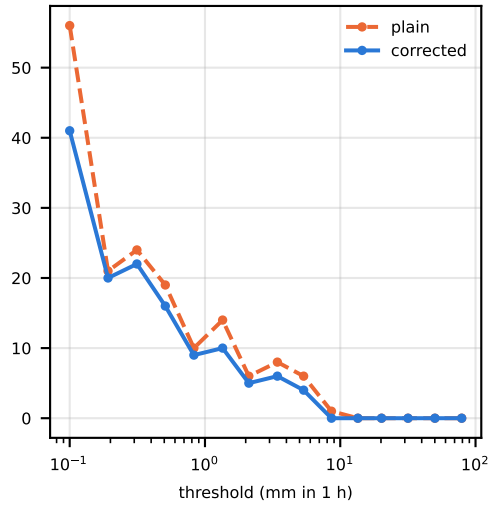
advection-corrected (max 8.6 mm)



corrected - plain (mm)

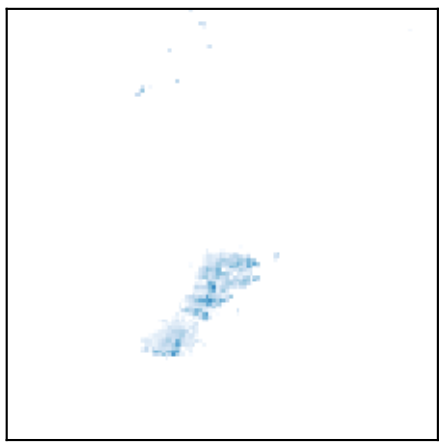


Euler characteristic χ

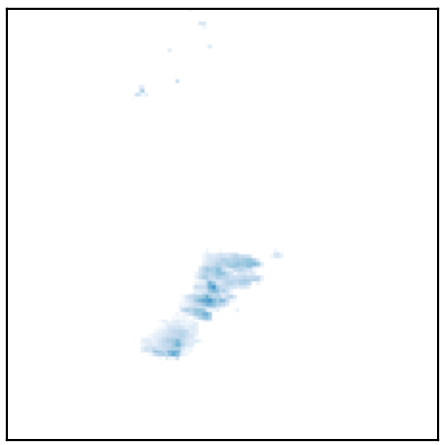


event
2016-06-04 08:00
51.3N 2.2E
elvira_friederike_201605

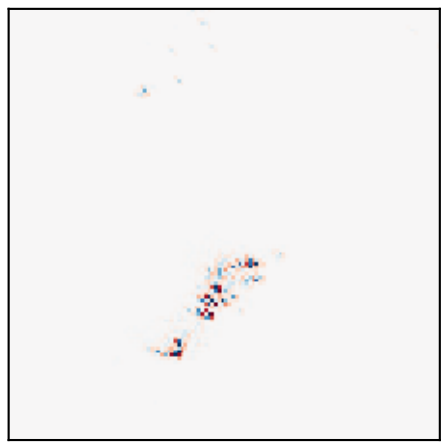
plain sum (max 8.1 mm)



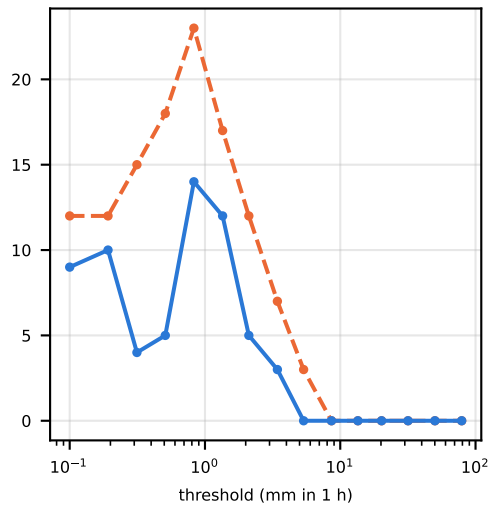
advection-corrected (max 5.4 mm)



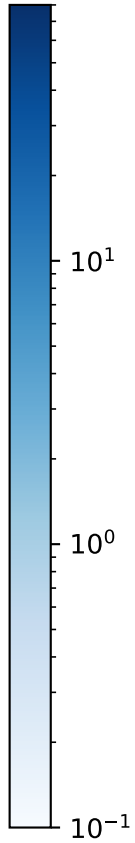
corrected - plain (mm)



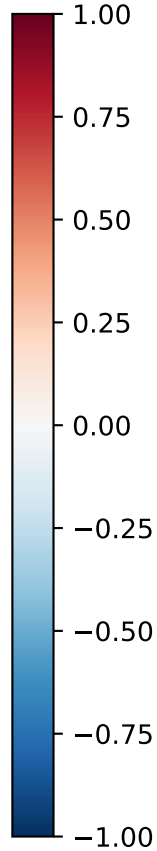
Euler characteristic χ



mm in 1 h (log)

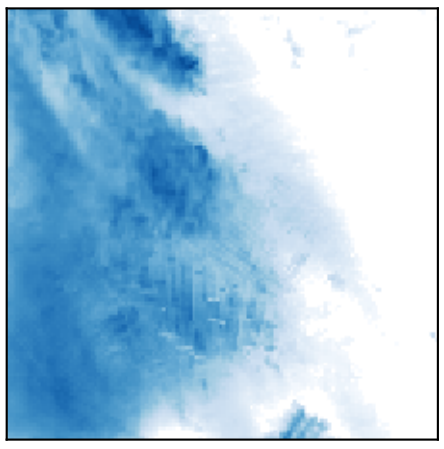


difference (mm)

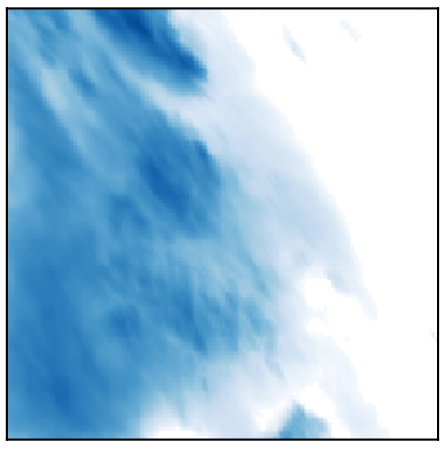


event
2022-08-17 21:00
44.6N 6.4E
med_derecho_20220818

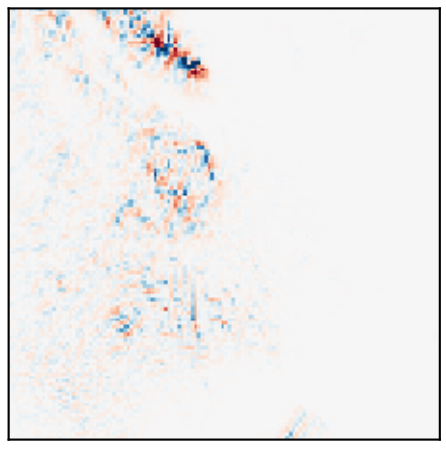
plain sum (max 46.4 mm)



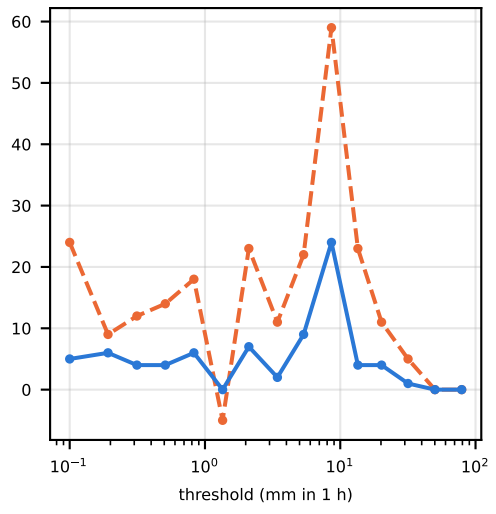
advection-corrected (max 40.1 mm)



corrected - plain (mm)

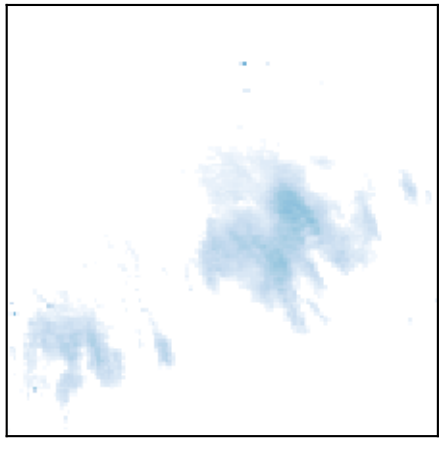


Euler characteristic χ

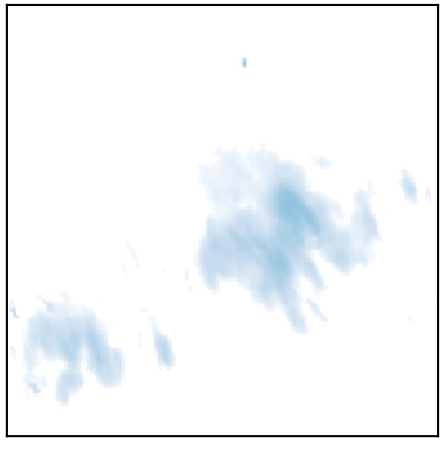


event
2016-06-01 09:00
49.3N 9.6E
elvira_friederike_201605

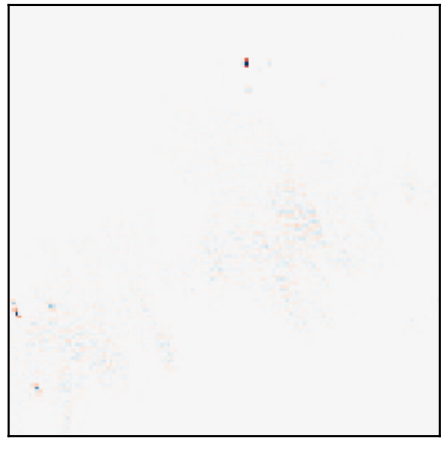
plain sum (max 2.5 mm)



advection-corrected (max 1.5 mm)



corrected - plain (mm)



Euler characteristic χ

Plain vs advection-corrected — samples 85-88 of 100

plain sum (max 3.2 mm)

advection-corrected (max 2.3 mm)

corrected - plain (mm)

Euler characteristic χ

event
2016-06-05 09:00
51.6N 9.6E
elvira_friederike_201605

plain sum (max 6.8 mm)

advection-corrected (max 5.3 mm)

corrected - plain (mm)

Euler characteristic χ

event
2014-06-09 06:00
49.2N 6.1E
pentecost_ela_20140609

mm in 1 h (log)

difference (mm)

plain sum (max 7.4 mm)

advection-corrected (max 3.7 mm)

corrected - plain (mm)

Euler characteristic χ

event
2022-08-18 13:00
46.8N 16.3E
med_derecho_20220818

plain sum (max 109.8 mm)

advection-corrected (max 68.4 mm)

corrected - plain (mm)

Euler characteristic χ

event
2023-07-24 19:00
46.9N 13.0E
chauxdefonds_italy_20230724

plain

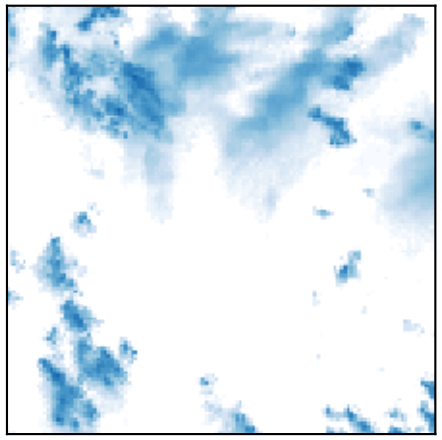
corrected

threshold (mm in 1 h)

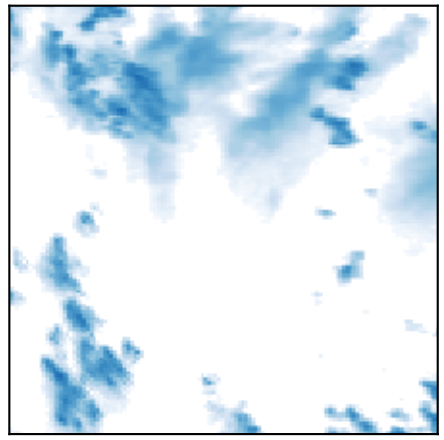
Plain vs advection-corrected — samples 89-92 of 100

event
2014-07-29 17:00
51.6N 9.6E
muenster_20140728

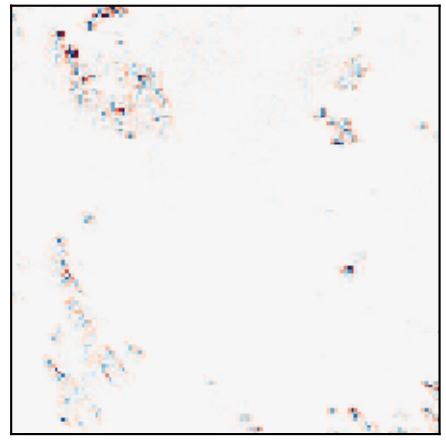
plain sum (max 24.6 mm)



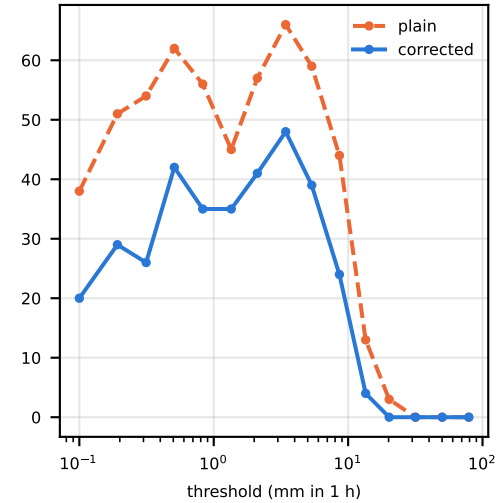
advection-corrected (max 18.7 mm)



corrected - plain (mm)

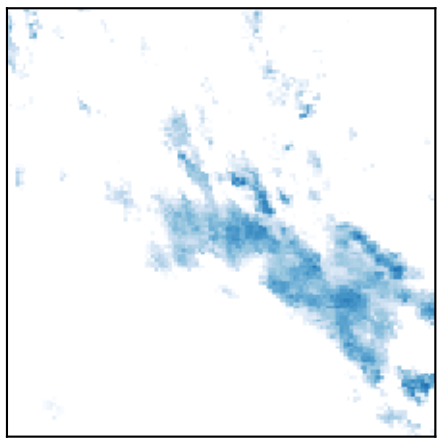


Euler characteristic χ

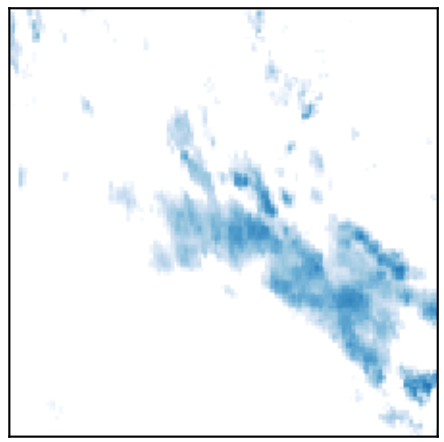


event
2013-06-01 13:00
46.8N 16.3E
ce_floods_201305

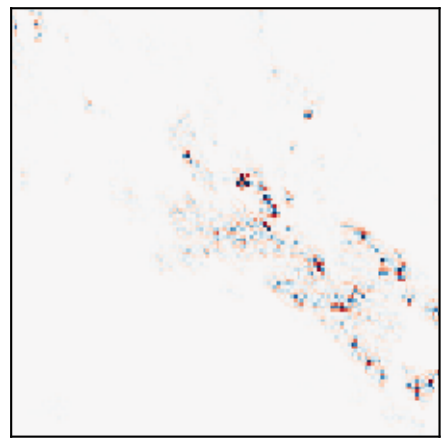
plain sum (max 15.1 mm)



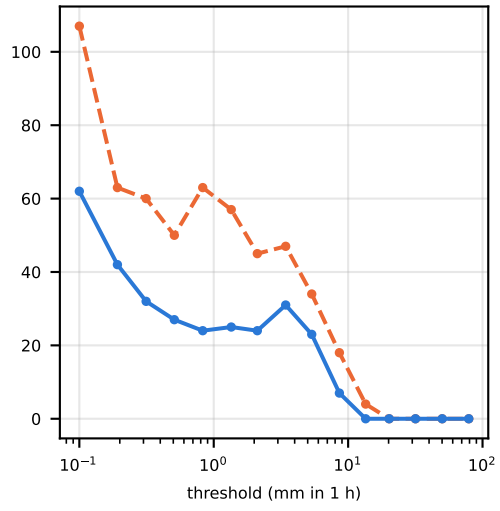
advection-corrected (max 13.5 mm)

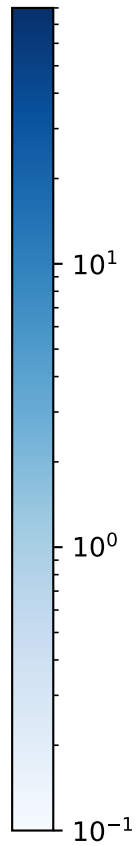


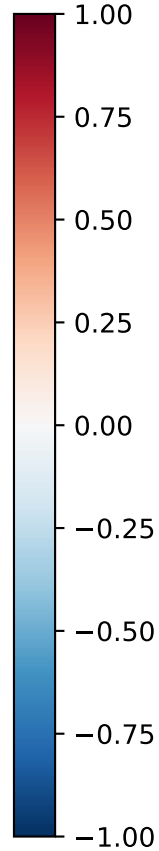
corrected - plain (mm)



Euler characteristic χ

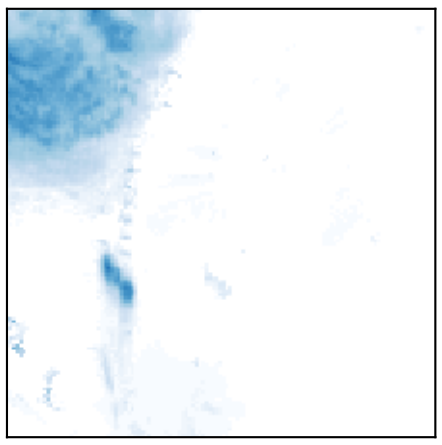




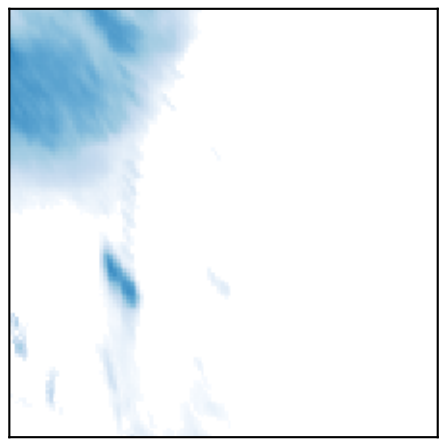


event
2013-07-27 14:00
53.9N 9.5E
andreas_20130728

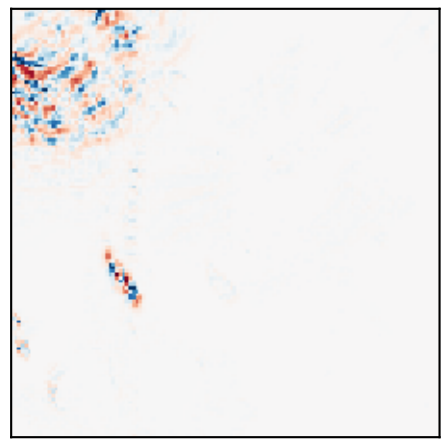
plain sum (max 14.2 mm)



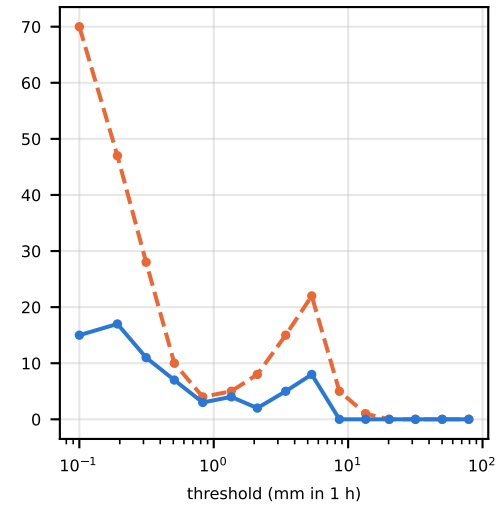
advection-corrected (max 7.6 mm)



corrected - plain (mm)

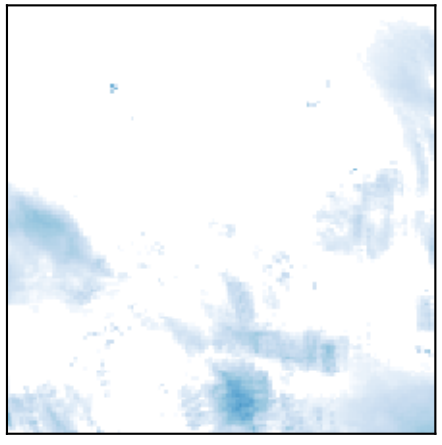


Euler characteristic χ

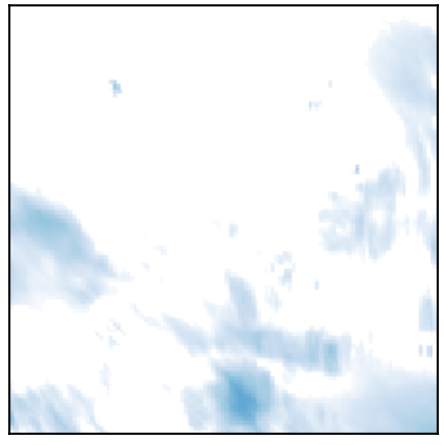


event
2016-06-01 15:00
49.2N 6.1E
elvira_friederike_201605

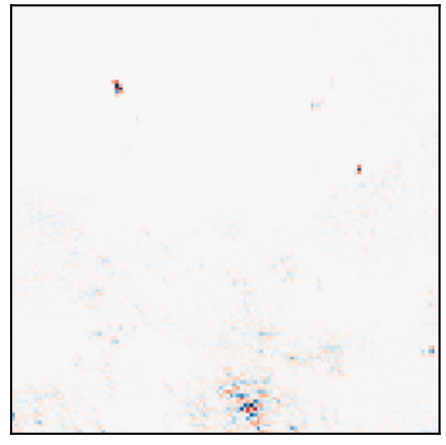
plain sum (max 5.2 mm)



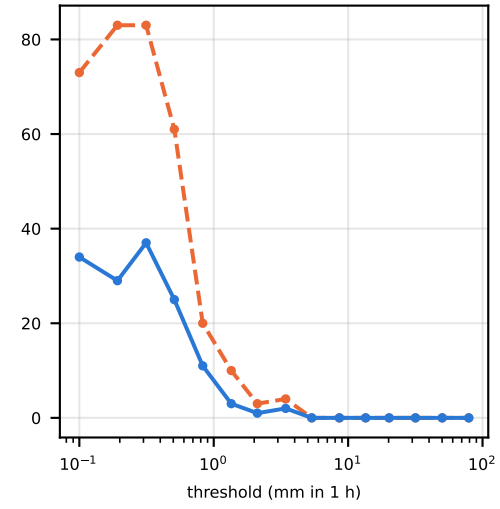
advection-corrected (max 3.7 mm)



corrected - plain (mm)



Euler characteristic χ



Plain vs advection-corrected — samples 93-96 of 100

plain sum (max 13.8 mm)

advection-corrected (max 11.1 mm)

corrected - plain (mm)

Euler characteristic χ

plain

corrected

threshold (mm in 1 h)

event
2024-05-31 18:00
46.9N 13.0E
s_germany_20240601

plain sum (max 1.6 mm)

advection-corrected (max 1.5 mm)

corrected - plain (mm)

Euler characteristic χ

threshold (mm in 1 h)

event
2016-06-03 02:00
51.5N 5.9E
elvira_friederike_201605

mm in 1 h (log)

difference (mm)

plain sum (max 3.8 mm)

advection-corrected (max 3.0 mm)

corrected - plain (mm)

Euler characteristic χ

threshold (mm in 1 h)

event
2022-08-18 08:00
44.4N 3.2E
med_derecho_20220818

plain sum (max 11.8 mm)

advection-corrected (max 11.8 mm)

corrected - plain (mm)

Euler characteristic χ

threshold (mm in 1 h)

event
2024-09-13 23:00
51.5N 13.3E
boris_20240913

Plain vs advection-corrected — samples 97-100 of 100

plain sum (max 1.4 mm)

advection-corrected (max 1.0 mm)

corrected - plain (mm)

Euler characteristic χ

plain

corrected

threshold (mm in 1 h)

event
2016-05-28 06:00
51.3N 2.2E
elvira_friederike_201605

plain sum (max 13.4 mm)

advection-corrected (max 10.3 mm)

corrected - plain (mm)

Euler characteristic χ

threshold (mm in 1 h)

event
2024-09-14 07:00
44.5N 16.0E
boris_20240913

mm in 1 h (log)

difference (mm)

plain sum (max 10.0 mm)

advection-corrected (max 3.6 mm)

corrected - plain (mm)

Euler characteristic χ

threshold (mm in 1 h)

event
2013-06-03 20:00
49.1N 16.6E
ce_floods_201305

plain sum (max 4.9 mm)

advection-corrected (max 3.5 mm)

corrected - plain (mm)

Euler characteristic χ

threshold (mm in 1 h)

event
2022-08-17 03:00
46.8N 16.3E
med_derecho_20220818